# Signatures

## What the inner product has been assuming, how to declare it differently, and what each declaration does

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# What This Session Covers

| | | |
|---|---|---|
| **I** | The matrix in the middle | the inner products you have already used this term, and what they share |
| **II** | The definition | declaring the square of each axis; lengths, similarities, and pictures |
| **III** | Polarization | how quadratic scores determine symmetric comparisons, and what they miss |
| **IV** | What a comparison preserves | which distinctions a learned comparison keeps; eigenvectors find the directions, the signature counts them |
| **V** | What a signature does | examples, organized by the three possible answers |
| **VI** | Where it goes next | the signature as a rule for multiplying vectors (Session 13a) |

Four in-class exercises, one at the end of each of Parts II through V. Every number in the deck is small enough to check by hand.

# Four Questions, and Where Each Is Answered

| **question** | **answer, in short** |
|---|---|
| Did I need a signature before? | Yes. Every inner product you have used had one; the dot product's was supplied for you. Part I |
| What is a signature? | For each axis, the declared value of its own square: $+1$, $-1$, or $0$. Part II |
| What is polarization? | The identity by which quadratic scores determine symmetric comparisons. Part III |
| What does a signature do? | It fixes which vectors have length zero, and which maps preserve length. Parts IV, V |

By the end you should be able to: write the dot product's signature; compute lengths and similarities under a declared signature; polarize a quadratic form by hand; read a signature off eigenvalues; and say what a $-1$ axis and a $0$ axis each make possible.

# Part I
# The Matrix in the Middle

the inner products you have already used

# The Dot Product, Written With a Matrix in the Middle

Four binary features, furry, barks, purrs, pet, and two words: dog $= (1,1,0,1)$, cat $= (1,0,1,1)$.

$$\text{dog} \cdot \text{cat} \;=\; \sum_i d_i c_i \;=\; 1+0+0+1 \;=\; 2, \qquad \cos(\text{dog}, \text{cat}) = \frac{2}{\sqrt{3}\sqrt{3}} = \tfrac{2}{3}$$

The same computation, with a matrix written where it has always silently been:

$$\text{dog} \cdot \text{cat} \;=\; \mathbf{d}^\top I\, \mathbf{c} \;=\; (1,1,0,1) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} = 2$$

In general $\mathbf{x}^\top M \mathbf{y} = \sum_i \sum_j x_i\, M_{ij}\, y_j$. With $M = I$ only the terms $i = j$ survive, and the sum of products comes back. Every similarity computed in this course so far has had the identity matrix sitting in that middle position. This session is about what happens when something else is put there.

# What the Identity Matrix Says

In $\mathbf{u}^\top M\mathbf{v} = \sum_i \sum_j u_i\, M_{ij}\, v_j$, each entry $M_{ij}$ weights one product, $u_i v_j$: coordinate $i$ of the first vector times coordinate $j$ of the second. The identity makes two statements:

| **entries** | **pair the coordinates** | **value** | **statement, for dog and cat** |
|---|---|---|---|
| diagonal, $M_{ii}$ | $u_i$ with $v_i$: one feature, lined up in both | $1$ | each feature counts, and counts like every other |
| off-diagonal, $M_{ij}$ | $u_i$ with $v_j$: different features | $0$ | no two features interact: barks never meets purrs |

Four features give sixteen products, one per cell of $M$. The identity keeps the four aligned ones, $d_1c_1 + d_2c_2 + d_3c_3 + d_4c_4$, and gives the twelve cross products weight $0$. With $\mathbf{e}_i$ the $i$th axis ($1$ in position $i$, $0$ elsewhere), $M_{ii} = \mathbf{e}_i^\top M\mathbf{e}_i$ is what $M$ says about axis $i$ alone.

## Two choices, not one

The second statement, independent axes, we will keep throughout. The first, every axis has square $1$, is the one we will open up. A **signature** is a different answer to the first statement, given axis by axis.

# Other Matrices in the Middle, Seen This Term

The identity is not the only matrix that has sat in that position:

| **inner product** | **matrix in the middle** | **what it measures** |
|---|---|---|
| dot product, cosine | $I$ | feature-by-feature agreement, all features alike |
| attention head, $\mathbf{x}^\top G\,\mathbf{y}$ | $G = W_Q^\top W_K$, learned | raw score of query $\mathbf{x}$ against key $\mathbf{y}$ (Session 11) |
| its symmetric half | $S = \frac{1}{2}(G + G^\top)$ | the part of the score unchanged when query and key swap (the bridge deck) |

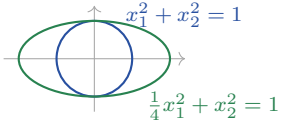
Session 10 put it this way: the matrix in the middle can be *learned* (the head), *assumed* (the identity), or *declared*. Row one is assumed, rows two and three learned; declaring is what this session adds.

All three share one thing: what a vector has with itself, $\mathbf{x}^\top M\mathbf{x}$, sees only the symmetric part of $M$ (Part III shows why). We call $Q(\mathbf{x}) = \mathbf{x}^\top M\mathbf{x}$ the **quadratic form** of $M$, and its value the **squared length** of $\mathbf{x}$ under $M$.

# Why You Already Had a Signature

Each matrix in the middle gives a quadratic form, with three numbers attached: how many independent directions have positive squared length, how many negative, how many zero. That triple is its **signature**; Part II defines it, Part IV computes it:

| **form** | $Q(\mathbf{x})$ in 2D | **signature** |
|---|---|---|
| dot product | $x_1^2 + x_2^2$ | $(2, 0, 0)$ |
| weights $\mathrm{diag}(\frac{1}{4}, 1)$ | $\frac{1}{4}x_1^2 + x_2^2$ | $(2, 0, 0)$ |
| the head's $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ | $(x_1 + x_2)^2$ | $(1, 0, 1)$ |




Two forms, one signature: every direction positive on both.

## The answer to the first question

You did need one; the identity supplied it, $(n, 0, 0)$. You never had to choose because every form you were handed was all-plus, except the head, which chose for itself and came out with a $0$. The signature is what a form keeps after the sizes are forgotten.

# Part II
# The Definition

declaring the square of each axis

# What a Declaration Is

Session 10 named three ways to supply the matrix in the middle of $\mathbf{x}^\top M \mathbf{y}$. They differ in who chooses $M$, when, and what you know about it:

| | **who chooses $M$** | **what $M$ can be** | **what you know about it** |
|---|---|---|---|
| **assumed** | no one: a default | $I$ | every feature counts alike; nobody decided it |
| **learned** | training, from data | any $d \times d$ matrix | what it does on data; why, only by analysis |
| **declared** | the modeler, in advance | one sign per axis, $+1$, $-1$, $0$ | everything, before any data |

**Definition**

A **declaration** is a choice, made before any data are seen, of independent axes and of the square of each one: $+1$, $-1$, or $0$.

The dot product is a declaration too, every axis $+1$, used all term without being stated. This session states it, then changes it.

# What a Declaration Does

One sign per axis; three things follow, each in its own part:

| the declaration fixes | how | where |
|---|---|---|
| every length | $Q(\mathbf{x}) = \sum_i (\mathbf{e}_i \cdot \mathbf{e}_i)\, x_i^2$: signed squares | Part II |
| every similarity | polarization: quadratic scores fix symmetric comparisons | Part III |
| which changes keep length | those holding $Q$ constant; one family per sign pattern | Part V |

**The third row, in two dimensions.** The signs decide what a length-preserving change must hold constant, and so the shape it moves along:

| signs | held constant | the set $Q = 1$ | a change that keeps $Q$ |
|---|---|---|---|
| $(+1, +1)$ | $x^2 + y^2$ | a circle: a closed curve | around the circle; can come back |
| $(+1, -1)$ | $x^2 - y^2$ | a hyperbola: open branches | along a branch; no wrap-around |

# Why a Declaration Matters

- **It states in advance which changes a meaning space allows.** Choosing mixed signs permits nonperiodic, length-preserving changes (boosts, Part V). Repeated positive boosts model cumulative change without wrap-around; rotations, available with all axes $+1$, model cyclic change.
- **It permits; it does not decide.** The signature permits a family of transformations; choosing the transformation and its direction specifies the semantic behavior. A negative axis does not by itself impose one-way meaning: a boost can be undone by its inverse. Part V's seasons choose a quarter turn on a circle; its *better for cutting* chooses a repeated boost on a hyperbola.
- **It can be tested.** Stated before the data, it can be wrong, and an experiment can show it; a learned matrix fits the data without any such claim.
- **It is all the algebra needs.** Session 13a's product of vectors needs nothing else (Part VI).

# One Axis: Declaring Its Square

Keep the axes independent, $\mathbf{e}_i \cdot \mathbf{e}_j = 0$ for $i \neq j$. Then for a single axis $\mathbf{e}$ one quantity remains to be fixed, its own square $\mathbf{e} \cdot \mathbf{e}$, and we allow three values. For a vector $x\,\mathbf{e}$ along that axis, the squared length is $Q(x\,\mathbf{e}) = x^2\,(\mathbf{e} \cdot \mathbf{e})$:

| **declared** | **we say the axis** | $Q(x\,\mathbf{e})$ | **with** $x = 3$ |
|---|---|---|---|
| $\mathbf{e} \cdot \mathbf{e} = +1$ | counts | $+x^2$ | $9$ |
| $\mathbf{e} \cdot \mathbf{e} = -1$ | counts against | $-x^2$ | $-9$ |
| $\mathbf{e} \cdot \mathbf{e} = 0$ | does not count | $0$ | $0$ |

Nothing else is declared. The coefficient $3$ is an ordinary real number and multiplies as it always did. The axis is told one thing only: what its own square is.

With the dot product every axis was declared $+1$ without anyone saying so. The other two rows are new, and the rest of the session is about what they do.

# Why the Declared Square Is $+1$, $-1$, or $0$, and Nothing Else

Suppose an axis came with $\mathbf{e} \cdot \mathbf{e} = 3$. Replace it by the rescaled axis $\mathbf{e}' = \mathbf{e}/\sqrt{3}$:

$$\mathbf{e}' \cdot \mathbf{e}' = \tfrac{1}{\sqrt{3}} \cdot \tfrac{1}{\sqrt{3}} \, (\mathbf{e} \cdot \mathbf{e}) = \tfrac{1}{3} \cdot 3 = 1$$

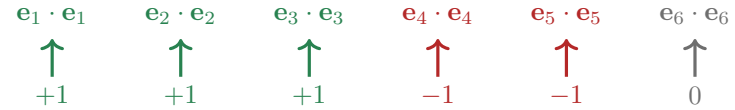
The same vector $x\,\mathbf{e}$ is now written $(\sqrt{3}\,x)\,\mathbf{e}'$, and its squared length is unchanged: $(\sqrt{3}x)^2 \cdot 1 = 3x^2$. Only the unit on the axis changed.

| **given** | **rescale by** | **becomes** |
|---|---|---|
| $\mathbf{e} \cdot \mathbf{e} = 3$ | $1/\sqrt{3}$ | $+1$ |
| $\mathbf{e} \cdot \mathbf{e} = -5$ | $1/\sqrt{5}$ | $-1$ |
| $\mathbf{e} \cdot \mathbf{e} = 0$ | anything | $0$ |

The size of the square is a choice of unit and can always be normalized away. The sign cannot: no real rescaling turns $-5$ into a positive number, and nothing turns $0$ into anything else. So three values are all that a declaration can distinguish.

# Several Axes, Each Declared: The Signature

Declare every axis and count the answers. Six axes, for instance:




$\Rightarrow\ (p, q, r) = (3, 2, 1)$

## Definition

Take $n$ independent axes, and declare the square of each to be $+1$, $-1$, or $0$. The **signature** is the triple $(p, q, r)$: $p$ axes with square $+1$, $q$ with square $-1$, $r$ with square $0$, and $p + q + r = n$.

| space | the axes answer | signature |
|---|---|---|
| word vectors in $\mathbb{R}^{300}$, with the cosine | all $+1$ | $(300, 0, 0)$ |
| the plane, with one axis counting against | $+1,\ -1$ | $(1, 1, 0)$ |
| the attention head's $S$, in its own axes (Part IV) | $+1,\ 0$ | $(1, 0, 1)$ |
| three axes, $D = \mathrm{diag}(1, -1, 0)$, used next | $+1,\ -1,\ 0$ | $(1, 1, 1)$ |

Part IV shows the counts do not depend on which axes are chosen.

# Generating Axes, and the Name $\mathrm{Cl}(p, q, r)$

## Generating axes

The **generating axes** are the independent axes $\mathbf{e}_1, \ldots, \mathbf{e}_n$ whose squares you declare. Every vector is built from them by adding and scaling, $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$; once vectors can be multiplied (Session 13a), everything else is built from them by multiplying too.

**Why the squares are the rules.** Products of generating axes are words like $\mathbf{e}_1\mathbf{e}_2$ or $\mathbf{e}_2\mathbf{e}_1\mathbf{e}_2$. Wherever an axis meets itself, $\mathbf{e}_i\mathbf{e}_i$, the declaration says what it becomes: $+1$, $-1$, or $0$. The signature is the rule book for those words, which is why it names the algebra.

**The name.** $\mathrm{Cl}(p, q, r)$ is the *Clifford algebra* with $p$ generating axes squaring to $+1$, $q$ to $-1$, $r$ to $0$ (W. K. Clifford, 1878; built in Session 13a). **To find it**: for a declaration, count the signs; for another symmetric matrix, find the axes where it is diagonal and count there (Part IV).

| the space | generating axes square to | algebra |
|---|---|---|
| $D = \mathrm{diag}(1, -1, 0)$, the next slides' example | $+1,\ -1,\ 0$ | $\mathrm{Cl}(1, 1, 1)$ |
| the six axes of the previous slide | $+1, +1, +1, -1, -1, 0$ | $\mathrm{Cl}(3, 2, 1)$ |
| word vectors in $\mathbb{R}^{300}$ with the cosine | all $+1$ | $\mathrm{Cl}(300, 0, 0)$ |

# The Signature as the Matrix in the Middle

Part I wrote every inner product as $\mathbf{x}^\top M \mathbf{y}$. A declaration is the special case where $M$ is diagonal with entries $+1$, $-1$, $0$:

$$\mathbf{x}\cdot\mathbf{y} \;=\; \mathbf{x}^\top D\,\mathbf{y}, \qquad D = \begin{pmatrix} \mathbf{e}_1\cdot\mathbf{e}_1 & & \\ & \ddots & \\ & & \mathbf{e}_n\cdot\mathbf{e}_n \end{pmatrix} \qquad \text{for example} \qquad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Multiplying out (the next slide does it term by term), only the $i = j$ terms survive, each carrying its sign:

$$\mathbf{x}\cdot\mathbf{y} = \sum_i (\mathbf{e}_i\cdot\mathbf{e}_i)\, x_i y_i, \qquad Q(\mathbf{x}) = \mathbf{x}\cdot\mathbf{x} = \sum_i (\mathbf{e}_i\cdot\mathbf{e}_i)\, x_i^2$$

With the example $D$, for *any* $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$: $\mathbf{x}\cdot\mathbf{y} = x_1y_1 - x_2y_2$ and $Q(\mathbf{x}) = x_1^2 - x_2^2$. The third coordinates appear in neither.

## Declared matrices and other matrices

A declaration is a *diagonal* matrix of signs; the dot product is $D = I$. Any other symmetric matrix, such as a head's $S$, becomes one in the right axes (Part IV), and its signature is the signs that come out.

# Multiplying Out $\mathbf{x}^\top D\,\mathbf{y}$, Every Term

Take any $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$, and $D = \mathrm{diag}(1, -1, 0)$. Two routes to the same answer.

**Route 1: matrix times vector, then dot product.**

$$D\mathbf{y} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 \\ 0 \cdot y_1 - 1 \cdot y_2 + 0 \cdot y_3 \\ 0 \cdot y_1 + 0 \cdot y_2 + 0 \cdot y_3 \end{pmatrix} = \begin{pmatrix} y_1 \\ -y_2 \\ 0 \end{pmatrix}, \quad \mathbf{x}^\top (D\mathbf{y}) = x_1 y_1 - x_2 y_2 + x_3 \cdot 0$$

**Route 2: all nine terms of the double sum $\sum_i \sum_j x_i\, D_{ij}\, y_j$, one per cell of $D$:**

| $x_i\, D_{ij}\, y_j$ | $j = 1$ | $j = 2$ | $j = 3$ |
|---|---|---|---|
| $i = 1$ | $x_1 \cdot (+1) \cdot y_1 = x_1 y_1$ | $x_1 \cdot 0 \cdot y_2 = 0$ | $x_1 \cdot 0 \cdot y_3 = 0$ |
| $i = 2$ | $x_2 \cdot 0 \cdot y_1 = 0$ | $x_2 \cdot (-1) \cdot y_2 = -x_2 y_2$ | $x_2 \cdot 0 \cdot y_3 = 0$ |
| $i = 3$ | $x_3 \cdot 0 \cdot y_1 = 0$ | $x_3 \cdot 0 \cdot y_2 = 0$ | $x_3 \cdot 0 \cdot y_3 = 0$ |

Add the nine cells: $\mathbf{x}^\top D\mathbf{y} = x_1 y_1 - x_2 y_2$. The six off-diagonal cells are $0$ because the axes are independent; the three diagonal cells are the aligned products $x_i y_i$, each multiplied by its axis's declared square. With $I$ in the middle the grid is the same except that all three diagonal weights are $+1$, which gives $x_1 y_1 + x_2 y_2 + x_3 y_3$, the dot product.

# The Same Two Vectors, Under $I$ and Under a Declaration

Take $\mathbf{u} = (2, 1, 3)$ and $\mathbf{v} = (1, 4, 2)$, and put two different matrices in the middle: $I$, and $D = \mathrm{diag}(1, -1, 0)$, the declaration $(+1, -1, 0)$. Multiply $\mathbf{v}$ by the matrix, then dot with $\mathbf{u}$:

$$\mathbf{u}^\top I\,\mathbf{v} = (2,1,3) \cdot \underbrace{(1,\ 4,\ 2)}_{I\mathbf{v}} = 12 \qquad \mathbf{u}^\top D\,\mathbf{v} = (2,1,3) \cdot \underbrace{(1,\ -4,\ 0)}_{D\mathbf{v}} = -2$$

The same computation, axis by axis. The aligned products $u_i v_i$ are the same in both; only the weight on each changes:

| | **axis 1** | **axis 2** | **axis 3** | **sum** |
|---|---|---|---|---|
| aligned product $u_i v_i$ | $2 \cdot 1 = 2$ | $1 \cdot 4 = 4$ | $3 \cdot 2 = 6$ | |
| under $I$: weight $\times$ product | $(+1) \cdot 2 = 2$ | $(+1) \cdot 4 = 4$ | $(+1) \cdot 6 = 6$ | $\mathbf{u}^\top I\mathbf{v} = 12$ |
| under $D$: weight $\times$ product | $(+1) \cdot 2 = 2$ | $(-1) \cdot 4 = -4$ | $0 \cdot 6 = 0$ | $\mathbf{u}^\top D\mathbf{v} = -2$ |

Axis 1 counts as before. Axis 2 counts against: agreeing there now lowers the similarity by exactly what it used to add. Axis 3 does not count: the largest agreement of the three, 6, is not seen at all. Lengths change the same way: $Q(\mathbf{u})$ is $4 + 1 + 9 = 14$ under $I$ but $4 - 1 + 0 = 3$ under $D$; $Q(\mathbf{v})$ is 21 under $I$ but $1 - 16 = -15$ under $D$.

# Length Under a Signature: A Worked Example

Three axes declared $(+1, -1, 0)$, so $Q(\mathbf{x}) = x_1^2 - x_2^2$:

| **vector** | $Q$ | | **reading** |
|---|---|---|---|
| $\mathbf{x} = (3,\ 1,\ 5)$ | $9 - 1 + 0$ | $= 8$ | positive, as the dot product would give |
| $\mathbf{y} = (1,\ 1,\ 7)$ | $1 - 1 + 0$ | $= 0$ | a nonzero vector of squared length zero |
| $\mathbf{z} = (1,\ 2,\ 0)$ | $1 - 4$ | $= -3$ | a negative squared length |
| $\mathbf{w} = (0,\ 0,\ 4)$ | $0$ | $= 0$ | entirely along the axis that does not count |

Three things the dot product never produced, one from each kind of axis: a negative squared length, a nonzero vector of length zero, and a coordinate that is written down and changes nothing. The third coordinate, $5$, $7$, or $4$, could be anything.

The word *length* is kept for continuity with the dot product. Under a signature $Q(\mathbf{x})$ is a signed bookkeeping quantity; it is not the length of an arrow on the page, and it need not be positive.

# Similarity Under a Signature: The Same Vectors

Under $(+1, -1, 0)$ the similarity is $\mathbf{x} \cdot \mathbf{y} = x_1 y_1 - x_2 y_2$, the same signs in front of the products. With $\mathbf{x} = (3, 1, 5)$, $\mathbf{y} = (1, 1, 7)$, $\mathbf{z} = (1, 2, 0)$, $\mathbf{w} = (0, 0, 4)$ as before:

| pair | computation | |
|---|---|---|
| $\mathbf{x} \cdot \mathbf{y}$ | $3 \cdot 1 - 1 \cdot 1 + 0 \cdot 5 \cdot 7$ | $= 2$ |
| $\mathbf{x} \cdot \mathbf{z}$ | $3 \cdot 1 - 1 \cdot 2$ | $= 1$ |
| $\mathbf{y} \cdot \mathbf{z}$ | $1 \cdot 1 - 1 \cdot 2$ | $= -1$ |
| $\mathbf{x} \cdot \mathbf{w}$ | $0$ | $= 0$, whatever $\mathbf{w}$ holds |

**First:** $\mathbf{y}$ has squared length $0$ but similarity $2$ with $\mathbf{x}$ and $-1$ with $\mathbf{z}$. Length zero does not mean similar to nothing. **Second:** the zero axis is invisible to every comparison. Two vectors that differ only along it have the same similarity to everything, and so are, for this inner product, the same vector.

A question to hold until Part III: the declaration fixed only the squares $\mathbf{e}_i \cdot \mathbf{e}_i$. Where did this formula for $\mathbf{x} \cdot \mathbf{y}$ come from, and could a different formula have the same lengths?

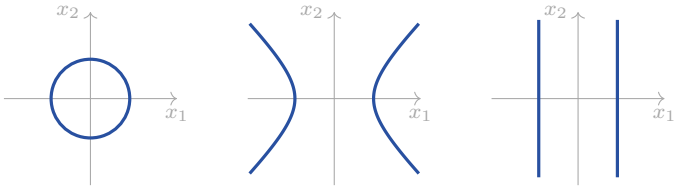
# Vocabulary

| term | meaning | example, three axes $(+1, -1, 0)$ |
|---|---|---|
| **null vector** | a nonzero vector with $Q = 0$ | $(1, 1, 7)$; also $(1, 1, 0)$ and $(0, 0, 4)$ |
| **null axis** | an axis declared $0$ | the third axis |
| **degenerate** signature | $r > 0$: at least one null axis | $(1, 1, 1)$; not $(1, 1, 0)$ on two axes |
| **definite** signature | all axes the same sign, no zeros | the dot product |
| **indefinite** signature | some $+1$ and some $-1$ | the plane with $(1, 1, 0)$ |

A null vector is not the zero vector: it has coordinates, a direction, and nonzero similarities; it is invisible to length alone. A definite signature has no null vectors; every other signature has them. A null *vector* arises in two ways: along a null *axis*, like $(0, 0, 4)$, or by cancellation between a $+1$ and a $-1$ axis, like $(1, 1, 0)$. The next two slides draw both cases.

# The Vectors of Squared Length 1, for Each Signature of the Plane

Draw every vector with $Q = 1$. For the dot product that is the unit circle.



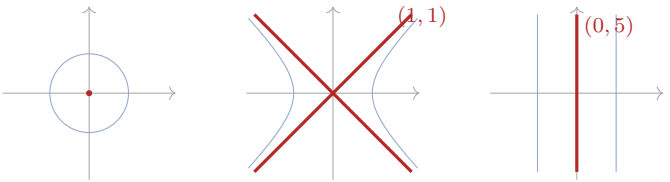

$(+1,+1)$: $x_1^2 + x_2^2 = 1$ $(+1,-1)$: $x_1^2 - x_2^2 = 1$ $(+1,\ 0)$: $x_1^2 = 1$

| **axes declared** | **the set $Q = 1$** | **how it reads** |
|---|---|---|
| $(+1,+1)$ | a circle | a unit vector in every direction, all alike |
| $(+1,-1)$ | a hyperbola, two branches | only directions nearer $x_1$ reach length 1 |
| $(+1,\ 0)$ | two parallel lines | length depends on $x_1$ alone; $x_2$ is free |

The shape of this set is the declaration, drawn. Part V shows that it is also the path along which length-preserving maps move a vector.

# The Vectors of Squared Length 0, for Each Signature of the Plane

Now draw every vector with $Q = 0$, in red. For the dot product it is a single point, the origin:




$(+1, +1)$: only $\mathbf{0}$ $(+1, -1)$: the two diagonals $(+1, 0)$: the whole $x_2$-axis

Check: under $(+1, -1)$, $Q(1, 1) = 1 - 1 = 0$, and so is every multiple of $(1, 1)$ or $(1, -1)$; under $(+1, 0)$, $Q(0, 5) = 0$, and so is every vector on the $x_2$-axis.

The middle picture shows the two ways a null vector arises, side by side with the hyperbola: the red diagonals are the hyperbola's asymptotes, the lines its branches approach and never reach. Null vectors are where the $+1$ axis and the $-1$ axis cancel exactly. In the right picture the null set is simply the axis that was declared $0$.

# In-Class Exercise 1

1. Three axes are declared $(+1, -1, 0)$. Find the squared lengths of $(2, 1, 9)$, $(1, 1, 1)$ and $(0, 3, 0)$. Which are null?
2. Under the same declaration, compute the similarities $(2, 1, 9) \cdot (1, 1, 1)$ and $(1, 1, 1) \cdot (0, 3, 0)$.
3. Write the matrix $D$ for the signature $(2, 1, 0)$ on three axes, with the $-1$ on the third axis, and compute $Q(1, 1, 1)$ and $Q(1, 1, \sqrt{2})$.
4. $\star$ Both axes of the plane are declared $-1$, signature $(0, 2, 0)$. Describe the set $Q = 1$ and the set $Q = -1$. Which of the two is the picture to draw?

Six minutes.

# In-Class Exercise 1: Worked Solution, Item 1

**1.** Under $(+1, -1, 0)$, $Q(\mathbf{x}) = (+1)\,x_1^2 + (-1)\,x_2^2 + 0 \cdot x_3^2$. Square each coordinate, multiply by its axis's sign, add:

| $\mathbf{x}$ | $x_1^2$ | $x_2^2$ | $x_3^2$ | $\times(+1)$ | $\times(-1)$ | $\times 0$ | $Q(\mathbf{x})$ | |
|---|---|---|---|---|---|---|---|---|
| $(2, 1, 9)$ | 4 | 1 | 81 | 4 | $-1$ | 0 | 3 | |
| $(1, 1, 1)$ | 1 | 1 | 1 | 1 | $-1$ | 0 | 0 | null |
| $(0, 3, 0)$ | 0 | 9 | 0 | 0 | $-9$ | 0 | $-9$ | |

**Which are null?** Only $(1, 1, 1)$. It is not the zero vector, but its $+1$ contribution and its $-1$ contribution cancel exactly. $(0, 3, 0)$ is not null: its length is negative, $-9$, because it lies entirely along the axis that counts against.

**What the zero axis did.** The $81$ from $x_3 = 9$ is the largest square on the slide, and the declaration discards it. With the dot product the three lengths would be $4 + 1 + 81 = 86$, $3$, and $9$.

# In-Class Exercise 1: Worked Solution, Item 2

**2.** Under the same declaration, $\mathbf{x} \cdot \mathbf{y} = (+1)\, x_1 y_1 + (-1)\, x_2 y_2 + 0 \cdot x_3 y_3$. Form the aligned products, weight them, add:

| pair | $x_1 y_1$ | $x_2 y_2$ | $x_3 y_3$ | $\times(+1)$ | $\times(-1)$ | $\times 0$ | $\mathbf{x} \cdot \mathbf{y}$ |
|---|---|---|---|---|---|---|---|
| $(2,1,9) \cdot (1,1,1)$ | 2 | 1 | 9 | 2 | $-1$ | 0 | 1 |
| $(1,1,1) \cdot (0,3,0)$ | 0 | 3 | 0 | 0 | $-3$ | 0 | $-3$ |

**Check the first with the matrix**, as on the multiplying-out slide: $D\mathbf{y} = D(1,1,1) = (1,-1,0)$, and $(2,1,9) \cdot (1,-1,0) = 2 - 1 + 0 = 1$.

**What it shows.** The null vector $(1,1,1)$ has length $0$ but similarities $1$ and $-3$ with the other two: null means invisible to its own length, not to comparison. With the dot product the two similarities would be $2 + 1 + 9 = 12$ and $0 + 3 + 0 = 3$.

# In-Class Exercise 1: Worked Solution, Item 3

**3.** Signature $(2,1,0)$ on three axes: two axes square to $+1$, one to $-1$, none to $0$. With the $-1$ on the third axis,

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad Q(\mathbf{x}) = x_1^2 + x_2^2 - x_3^2, \qquad \begin{aligned} Q(1,1,1) &= 1+1-1 = 1 \\ Q(1,1,\sqrt{2}) &= 1+1-2 = 0, \text{ null} \end{aligned}$$

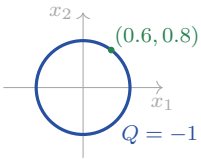
**Where all the null vectors are.** $Q = 0$ exactly when $x_3^2 = x_1^2 + x_2^2$: the third coordinate equals, up to sign, the ordinary length of the first two. Another one with whole numbers is $(3,4,5)$: $9 + 16 - 25 = 0$. At each height $x_3 = c$ the null vectors form a circle of radius $|c|$, so together they form a **cone**, the three-axis version of the two diagonals in the plane.

| $\mathbf{x}$ | $x_1^2 + x_2^2$ | $x_3^2$ | $Q = x_1^2 + x_2^2 - x_3^2$ | |
|---|---|---|---|---|
| $(1,1,1)$ | 2 | 1 | 1 | positive |
| $(1,1,\sqrt{2})$ | 2 | 2 | 0 | null |
| $(3,4,5)$ | 25 | 25 | 0 | null |
| $(1,0,2)$ | 1 | 4 | $-3$ | negative |

# In-Class Exercise 1: Worked Solution, Item 4

**4.** ⋆ Both axes $-1$, signature $(0, 2, 0)$: $Q(\mathbf{x}) = -x_1^2 - x_2^2 = -(x_1^2 + x_2^2)$.

- $x_1^2 + x_2^2 \geq 0$ for every vector, so $Q \leq 0$ always. $Q = 1$ would need $x_1^2 + x_2^2 = -1$: no vector does it, and the set is empty.
- $Q = -1$ means $x_1^2 + x_2^2 = 1$: the unit circle. Check: $Q(1, 0) = -1$; $Q(0.6,\ 0.8) = -0.36 - 0.64 = -1$.
- So draw the circle and label it $Q = -1$. Signature $(0, 2, 0)$ is the dot product with every length negated: the same shape, the opposite sign, and no null vectors but $\mathbf{0}$.




The lesson of item 4: flipping *every* sign changes nothing about the shape. What is new in a signature comes from *mixing* signs, as in item 3, or from a zero, as in item 1.

# Part III
# Polarization

how quadratic scores determine symmetric comparisons

# How Geometry Determines a Comparison Between Meanings

**The goal.** We represent meanings as vectors so that we can *compute relationships between them*. Vectors for dog, cat, democracy are representations; they do not say whether two are alike. That needs a rule for comparing them.

**The rule.** In embedding models it is the dot product or the cosine; underneath both is a **symmetric bilinear comparison** $B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top M \mathbf{y}$, $M$ symmetric. Coordinates supply the representations; $B$ decides how each feature contributes. Whether $B$ captures a useful semantic relation depends on how both were designed or learned.

**Single vectors.** Each vector also has a score of its own, its **quadratic score** $Q(\mathbf{x}) = B(\mathbf{x}, \mathbf{x})$: the comparison of a vector with itself. For the dot product, $Q$ is the squared length.

**The question for Part III.** A comparison $B$ takes two vectors; a score $Q$ takes one. $Q$ is built from $B$. Can we go the other way, and recover every comparison $B(\mathbf{x}, \mathbf{y})$ from scores of single vectors alone? If so, then declaring the scores, which is what a signature does, declares every comparison. The next slide shows that the answer is yes: **quadratic scores determine symmetric bilinear comparisons**.

**One assumption.** For a declaration, the generating axes are *assumed orthogonal*, $\mathbf{e}_i \cdot \mathbf{e}_j = 0$ for $i \neq j$; without that, the squares of the axes alone would not fix cross-axis comparisons.

# Where the Polarization Formula Comes From

**In ordinary arithmetic first.** $(a+b)^2 = a^2 + 2ab + b^2$, so $ab = \frac{1}{2}\big[(a+b)^2 - a^2 - b^2\big]$: a product recovered from squares. With $a = 3$, $b = 4$: $\frac{1}{2}(49 - 9 - 16) = 12$.

**The same move for vectors.** Score the sum $\mathbf{x} + \mathbf{y}$ and expand, one step per line:

| | line | what justifies it |
|---|---|---|
| (a) | $Q(\mathbf{x}+\mathbf{y}) = B(\mathbf{x}+\mathbf{y},\ \mathbf{x}+\mathbf{y})$ | definition of $Q$ |
| (b) | $= B(\mathbf{x},\mathbf{x}) + B(\mathbf{x},\mathbf{y}) + B(\mathbf{y},\mathbf{x}) + B(\mathbf{y},\mathbf{y})$ | distribute in each slot, as $(a+b)(c+d)$ |
| (c) | $= Q(\mathbf{x}) + 2\,B(\mathbf{x},\mathbf{y}) + Q(\mathbf{y})$ | $B(\mathbf{y},\mathbf{x}) = B(\mathbf{x},\mathbf{y})$: $M$ symmetric |
| (d) | $B(\mathbf{x},\mathbf{y}) = \frac{1}{2}\big[\,Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})\,\big]$ | solve (c) for $B(\mathbf{x},\mathbf{y})$ |

Line (d) is **polarization**. **Check**, dot product, $\mathbf{x} = (3,1)$, $\mathbf{y} = (1,2)$: $Q(\mathbf{x}) = 10$, $Q(\mathbf{y}) = 5$, $Q(4,3) = 25$, and $\frac{1}{2}(25 - 10 - 5) = 5 = 3 \cdot 1 + 1 \cdot 2$.

**In words:** score the sum of the two representations; subtract the score each gets on its own; what remains is twice their interaction. The sum $\mathbf{x} + \mathbf{y}$ is an algebraic probe; it need not represent a phrase made from the two words.

# Polarization With dog and cat

Four binary features, furry, barks, purrs, pet, and the dot product as the comparison $B$. Score each word, then score the probe dog + cat:

| | **vector** | **quadratic score $Q$** |
|---|---|---|
| dog | $(1,1,0,1)$ | $1+1+0+1=3$ |
| cat | $(1,0,1,1)$ | $1+0+1+1=3$ |
| the probe dog + cat | $(2,1,1,2)$ | $4+1+1+4=10$ |

**Polarize**: $B(\text{dog},\text{cat}) = \frac{1}{2}(10-3-3) = 2$, and directly $1+0+0+1=2$.

**Where the $4$ came from.** The probe scores $10$, not $3+3=6$. Shared features, furry and pet, become $2$ in the probe, and $2^2=4$ exceeds $1^2+1^2=2$: each shared feature adds $2$ beyond the separate scores. Unshared features, barks and purrs, stay at $1$ and add nothing beyond them. The interaction is the overlap, read off from scores alone.

**Why it matters.** Deciding how single representations are scored already commits a model to its comparisons, which can be tested against human similarity ratings. Since dog + cat = cat + dog, quadratic scores can determine only symmetric comparisons.

# Symmetric Comparisons and Ordered Relations

Polarization recovers only the part of a comparison that is the same in both directions. Similarity and ordered relations ask different things of a representation:

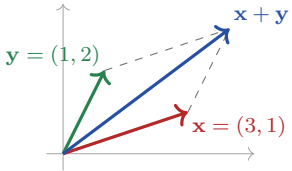
| **relation** | **example** | **the same both ways?** |
|---|---|---|
| similarity, relatedness | dog and cat | yes: within reach of $Q$ |
| synonymy | sofa and couch | yes |
| co-hyponymy | oak and maple, both trees | yes |
| hypernymy | a dog is an animal | no: an animal need not be a dog |
| entailment | *run* entails *move* | no |
| argument roles | *dog bites man* / *man bites dog* | no: same words, opposite events |
| attention | token 1 queries token 2, or the reverse | no: $24$ against $8$ (Session 11) |

Below the line, much of compositional semantics, lies outside what quadratic scores determine. Geometry fixes a consistent symmetric comparison; directional relationships need additional structure, which Session 13a supplies.

# Polarization With the Dot Product

The derivation once more, for the dot product, with the picture behind it:

$$\begin{aligned} Q(\mathbf{x}+\mathbf{y}) &= (\mathbf{x}+\mathbf{y})\cdot(\mathbf{x}+\mathbf{y}) \\ &= \mathbf{x}\cdot\mathbf{x}+\mathbf{x}\cdot\mathbf{y}+\mathbf{y}\cdot\mathbf{x}+\mathbf{y}\cdot\mathbf{y} \\ &= Q(\mathbf{x})+2\,\mathbf{x}\cdot\mathbf{y}+Q(\mathbf{y}) \end{aligned}$$




Solve for the similarity:

$$\boxed{\mathbf{x}\cdot\mathbf{y} \;=\; \tfrac{1}{2}\big[\,Q(\mathbf{x}+\mathbf{y})-Q(\mathbf{x})-Q(\mathbf{y})\,\big]}$$

The excess of the diagonal's squared length over the two sides', $25-10-5=10$, is twice the similarity: the law of cosines, $|\mathbf{x}+\mathbf{y}|^2=|\mathbf{x}|^2+|\mathbf{y}|^2+2|\mathbf{x}||\mathbf{y}|\cos\theta$, with the angle inside $\mathbf{x}\cdot\mathbf{y}$.

**Numbers.** $\mathbf{x}=(3,1)$, $\mathbf{y}=(1,2)$: $Q(\mathbf{x})=10$, $Q(\mathbf{y})=5$, $\mathbf{x}+\mathbf{y}=(4,3)$, $Q(4,3)=25$. So $\mathbf{x}\cdot\mathbf{y}=\frac{1}{2}(25-10-5)=5$, and directly $3\cdot1+1\cdot2=5$.

The identity is called **polarization**. Nothing in the derivation used the value $1$ for the squares of the axes, which is the point of the next slide.

# Polarization Under a Signature

Repeat the derivation under $(+1,-1,0)$ with the vectors of Part II, $\mathbf{x}=(3,1,5)$ and $\mathbf{y}=(1,1,7)$. The three lines are the same; only the lengths change:

| **quantity** | **computation under** $(+1,-1,0)$ | |
|---|---|---|
| $Q(\mathbf{x})$ | $9-1+0$ | $=8$ |
| $Q(\mathbf{y})$ | $1-1+0$ | $=0$ |
| $Q(\mathbf{x}+\mathbf{y})=Q(4,2,12)$ | $16-4+0$ | $=12$ |
| $\mathbf{x}\cdot\mathbf{y}=\frac{1}{2}[Q(\mathbf{x}+\mathbf{y})-Q(\mathbf{x})-Q(\mathbf{y})]$ | $\frac{1}{2}(12-8-0)$ | $=2$ |
| direct, from Part II | $3\cdot 1-1\cdot 1$ | $=2$ |

The two agree, and must: $\sum_i(\mathbf{e}_i\cdot\mathbf{e}_i)\,x_iy_i$ is the only symmetric bilinear function with the declared lengths. Declaring the squares of the axes declares every similarity, which answers the question held over from Part II.

A second form of the identity uses a difference: $\mathbf{x}\cdot\mathbf{y}=\frac{1}{4}[Q(\mathbf{x}+\mathbf{y})-Q(\mathbf{x}-\mathbf{y})]$. Here $\mathbf{x}-\mathbf{y}=(2,0,-2)$, $Q=4$, and $\frac{1}{4}(12-4)=2$ again: the similarity is a difference of two lengths, the bridge deck's move once more.

# Polarizing a Matrix That Is Not Symmetric: Expanding $Q(\mathbf{x}+\mathbf{y})$

The attention head $G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ is not symmetric. Its bilinear form $\mathbf{x}^\top G\mathbf{y}$ applies $G$ to the right-slot vector (the **key**), then dots with the left-slot vector (the **query**). Its quadratic score puts the *same* vector in both slots: $Q(\mathbf{x}) = \mathbf{x}^\top G\mathbf{x}$.

**Step 1: expand.** Put $\mathbf{x}+\mathbf{y}$ in both slots and distribute, as $(a+b)(c+d)$ gives four products:

$$Q(\mathbf{x}+\mathbf{y}) = (\mathbf{x}+\mathbf{y})^\top G(\mathbf{x}+\mathbf{y}) = \underbrace{\mathbf{x}^\top G\mathbf{x}}_{Q(\mathbf{x})} + \mathbf{x}^\top G\mathbf{y} + \mathbf{y}^\top G\mathbf{x} + \underbrace{\mathbf{y}^\top G\mathbf{y}}_{Q(\mathbf{y})}$$

| left slot (query) | right slot (key) | term | with $\mathbf{x}=(3,1)$, $\mathbf{y}=(1,3)$ |
|---|---|---|---|
| $\mathbf{x}$ | $\mathbf{x}$ | $\mathbf{x}^\top G\mathbf{x} = Q(\mathbf{x})$ | $(3,1)\cdot(5,1) = 16$ |
| $\mathbf{x}$ | $\mathbf{y}$ | $\mathbf{x}^\top G\mathbf{y}$ | $(3,1)\cdot(7,3) = 24$ |
| $\mathbf{y}$ | $\mathbf{x}$ | $\mathbf{y}^\top G\mathbf{x}$ | $(1,3)\cdot(5,1) = 8$ |
| $\mathbf{y}$ | $\mathbf{y}$ | $\mathbf{y}^\top G\mathbf{y} = Q(\mathbf{y})$ | $(1,3)\cdot(7,3) = 16$ |
| | total, $Q(\mathbf{x}+\mathbf{y}) = Q(4,4)$: | | $16+24+8+16 = 64$ |

**Step 2: subtract** $Q(\mathbf{x})$ and $Q(\mathbf{y})$. Two cross terms remain, one per assignment of the roles:

$$Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}) = \mathbf{x}^\top G\mathbf{y} + \mathbf{y}^\top G\mathbf{x} \qquad (64 - 16 - 16 = 24 + 8)$$

# Polarizing a Matrix That Is Not Symmetric: Combining the Cross Terms

From Steps 1–2, with $G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$: $\quad Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}) = \mathbf{x}^\top G\mathbf{y} + \mathbf{y}^\top G\mathbf{x}$.

**Step 3: put $\mathbf{x}$ on the left in both terms.** $\mathbf{y}^\top G\mathbf{x}$ is a single number, so it equals its own transpose; transposing a product reverses the order and transposes each factor:

$$\mathbf{y}^\top G\mathbf{x} = \left(\mathbf{y}^\top G\mathbf{x}\right)^\top = \mathbf{x}^\top G^\top \mathbf{y} \qquad \text{check: } G^\top = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right),\ G^\top(1,3) = (1,5),\ (3,1)\cdot(1,5) = 8$$

**Step 4: factor and halve.**

$$\tfrac{1}{2}\left[\mathbf{x}^\top G\mathbf{y} + \mathbf{x}^\top G^\top \mathbf{y}\right] = \mathbf{x}^\top \underbrace{\tfrac{1}{2}(G + G^\top)}_{S}\,\mathbf{y}, \qquad S = \tfrac{1}{2}\left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$$

**Step 5: see what disappeared.** Write $G = S + A$ with $A = \frac{1}{2}(G - G^\top) = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Because $A^\top = -A$, the two cross terms are

| | **symmetric part** | **antisymmetric part** | **with $\mathbf{x} = (3,1)$, $\mathbf{y} = (1,3)$** |
|---|---|---|---|
| $\mathbf{x}^\top G\mathbf{y}$ (x queries y) | $\mathbf{x}^\top S\mathbf{y}$ | $+\mathbf{x}^\top A\mathbf{y}$ | $24 = 16 + 8$ |
| $\mathbf{y}^\top G\mathbf{x}$ (y queries x) | $\mathbf{x}^\top S\mathbf{y}$ | $-\mathbf{x}^\top A\mathbf{y}$ | $8 = 16 - 8$ |
| half their sum | $\mathbf{x}^\top S\mathbf{y}$ | $0$ | $16$ |

Polarization averages the two role assignments; the antisymmetric part cancels, and only $S$ survives.

# Polarization Keeps Only the Symmetric Half

Steps 1–4 in one line, for the head $G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ with $Q(\mathbf{x}) = \mathbf{x}^\top G \mathbf{x}$:

$$\tfrac{1}{2}\big[Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})\big] = \tfrac{1}{2}\big[\mathbf{x}^\top G\mathbf{y} + \mathbf{y}^\top G\mathbf{x}\big] = \mathbf{x}^\top \underbrace{\tfrac{1}{2}(G + G^\top)}_{S} \mathbf{y}$$

The cross terms are the raw score with $\mathbf{x}$ as query and $\mathbf{y}$ as key, and the reverse; polarization averages them. The antisymmetric half $A = \frac{1}{2}(G - G^\top)$ is gone: its contributions $\mathbf{x}^\top A\mathbf{y}$ and $\mathbf{y}^\top A\mathbf{x} = -\mathbf{x}^\top A\mathbf{y}$ cancel.

| **with** $\mathbf{x}_1 = (3,1)$, $\mathbf{x}_2 = (1,3)$ | | | |
|---|---|---|---|
| $Q(\mathbf{x}_1)$, $Q(\mathbf{x}_2)$, $Q(\mathbf{x}_1 + \mathbf{x}_2)$ | 16, 16, $Q(4,4) = 64$ | | |
| polarized | $\frac{1}{2}(64 - 16 - 16)$ | $= 16$ | $= \mathbf{x}_1^\top S \mathbf{x}_2$ |
| the two raw scores | $\mathbf{x}_1^\top G\mathbf{x}_2 = 24$, $\mathbf{x}_2^\top G\mathbf{x}_1 = 8$ | average 16 | half-difference 8, lost |

## Consequence

A quadratic form remembers only the symmetric half of the matrix that defined it. A signature, read from a quadratic form (Part IV), is therefore a property of $S$ alone; the head's $A$ has no signature.

# What Polarization Recovers From an Attention Head, and What It Cannot

The head of Session 11, with $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and tokens $\mathbf{x}_1 = (3, 1)$, $\mathbf{x}_2 = (1, 3)$:

| what we want to inspect | what the mathematics provides |
|---|---|
| symmetric compatibility between token representations | $B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top S \mathbf{y}$ |
| each representation's self-score | $Q(\mathbf{x}) = \mathbf{x}^\top S \mathbf{x}$ |
| recovering symmetric compatibility from self-scores | polarization |
| different scores when query and key roles are reversed | the antisymmetric part $A$, which self-scores cannot recover |

**The numbers.** $Q(3,1) = 16$, $Q(1,3) = 16$, $Q(4,4) = 64$, so $B(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{2}(64 - 16 - 16) = 16$.

That recovers exactly the shared symmetric contribution to the raw scores $24 = 16 + 8$ and $8 = 16 - 8$.

It cannot recover the directional contributions $+8$ and $-8$.

The payoff: geometry determines a consistent symmetric comparison; who acts on whom needs additional structure.

# What Polarization Gives Later: The Product of Two Vectors

The bridge deck introduced a product of vectors $\mathbf{uv}$ whose symmetric half is the inner product and whose antisymmetric half is the wedge. Polarization is the reason the symmetric half is the inner product. Suppose only that every vector squares to its declared length, $\mathbf{vv} = Q(\mathbf{v})$, and that the product distributes over addition. Then

$$(\mathbf{u}+\mathbf{v})(\mathbf{u}+\mathbf{v}) = \mathbf{uu} + \mathbf{uv} + \mathbf{vu} + \mathbf{vv} = Q(\mathbf{u}) + \mathbf{uv} + \mathbf{vu} + Q(\mathbf{v})$$

and also $$(\mathbf{u}+\mathbf{v})(\mathbf{u}+\mathbf{v}) = Q(\mathbf{u}+\mathbf{v}) = Q(\mathbf{u}) + 2\,\mathbf{u}\cdot\mathbf{v} + Q(\mathbf{v}) \qquad \text{by polarization}$$

Subtract: $\boxed{\mathbf{uv} + \mathbf{vu} = 2\,\mathbf{u}\cdot\mathbf{v}}$ The symmetric half of the product is the declared similarity.

Two consequences, both used in the bridge deck without proof:

- For independent axes, $\mathbf{e}_i \cdot \mathbf{e}_j = 0$, so $\mathbf{e}_i\mathbf{e}_j = -\,\mathbf{e}_j\mathbf{e}_i$: different axes anticommute. This is the swap rule.
- For an axis with itself, $\mathbf{e}_i\mathbf{e}_i = \mathbf{e}_i \cdot \mathbf{e}_i$, the declared value: the signature is a multiplication rule. Part VI returns to this.

So the one declaration, squares of axes, fixes lengths (Part II), similarities (this part), and, once vectors multiply, the product of any two axes.

# In-Class Exercise 2

1. With the dot product, $\mathbf{x} = (2, 1)$ and $\mathbf{y} = (1, 4)$: compute $\mathbf{x} \cdot \mathbf{y}$ directly, then by polarization from $Q(\mathbf{x})$, $Q(\mathbf{y})$, $Q(\mathbf{x} + \mathbf{y})$.
2. The same two vectors under $(+1, -1)$: directly, then by polarization. (One of the three quadratic scores is negative; carry the sign.)
3. The head's self-score is $Q(\mathbf{x}) = (x_1 + x_2)^2$. Polarize it to find $\mathbf{x} \cdot \mathbf{y}$ for $\mathbf{x} = (3, 1)$ and $\mathbf{y} = (2, 5)$. Then compute the two scores $\mathbf{x}^\top G \mathbf{y}$ and $\mathbf{y}^\top G \mathbf{x}$ with $G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ and check that the polarized value is their average.
4. $\star$ Why must the inner product be symmetric for polarization to work? Find where the derivation on the dot-product slide uses $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.

Six minutes.

# In-Class Exercise 2: Answers, Items 1 and 2

$\mathbf{x} = (2, 1)$, $\mathbf{y} = (1, 4)$, probe $\mathbf{x} + \mathbf{y} = (3, 5)$. Each quadratic score: square each coordinate, weight it by its axis's declared square, add.

| | **1. dot product,** $(+1, +1)$ | **2. declared** $(+1, -1)$ |
|---|---|---|
| $Q(\mathbf{x}) = Q(2, 1)$ | $(+1) \cdot 2^2 + (+1) \cdot 1^2 = 4 + 1 = 5$ | $(+1) \cdot 2^2 + (-1) \cdot 1^2 = 4 - 1 = 3$ |
| $Q(\mathbf{y}) = Q(1, 4)$ | $(+1) \cdot 1^2 + (+1) \cdot 4^2 = 1 + 16 = 17$ | $(+1) \cdot 1^2 + (-1) \cdot 4^2 = 1 - 16 = -15$ |
| $Q(\mathbf{x} + \mathbf{y}) = Q(3, 5)$ | $(+1) \cdot 3^2 + (+1) \cdot 5^2 = 9 + 25 = 34$ | $(+1) \cdot 3^2 + (-1) \cdot 5^2 = 9 - 25 = -16$ |
| subtract the two singles | $34 - 5 - 17 = 12$ | $-16 - 3 - (-15) = -16 - 3 + 15 = -4$ |
| halve | $\frac{1}{2} \cdot 12 = 6$ | $\frac{1}{2} \cdot (-4) = -2$ |
| directly, axis by axis | $(+1) \cdot 2 \cdot 1 + (+1) \cdot 1 \cdot 4 = 2 + 4 = 6$ | $(+1) \cdot 2 \cdot 1 + (-1) \cdot 1 \cdot 4 = 2 - 4 = -2$ |

**Where the sign needs care.** In item 2, $Q(\mathbf{y}) = -15$, so subtracting it *adds* 15. Dropping that sign gives $\frac{1}{2}(-16 - 3 - 15) = -17$, which the direct $-2$ catches.

**What changed between the columns.** Only axis 2's declaration: its product $1 \cdot 4$ entered with $+$, then with $-$. Same vectors, comparison 6 became $-2$.

# In-Class Exercise 2: Answers, Item 3, Polarizing the Head's Self-Score

$\mathbf{x} = (3, 1)$, $\mathbf{y} = (2, 5)$, and the head's self-score $Q(\mathbf{x}) = (x_1 + x_2)^2$: add the two coordinates, then square.

| **step** | **vector** | $x_1 + x_2$ | $Q = (x_1 + x_2)^2$ |
|---|---|---|---|
| score $\mathbf{x}$ | $(3, 1)$ | $4$ | $16$ |
| score $\mathbf{y}$ | $(2, 5)$ | $7$ | $49$ |
| form the probe $\mathbf{x} + \mathbf{y}$ and score it | $(5, 6)$ | $11$ | $121$ |
| subtract the two singles | | | $121 - 16 - 49 = 56$ |
| halve | | | $\frac{1}{2} \cdot 56 = 28$ |

**Check against the matrix.** $(x_1 + x_2)^2 = x_1^2 + 2x_1x_2 + x_2^2$ has matrix $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$: $1$, $1$ on the diagonal for the squares, the cross coefficient $2$ split evenly between the off-diagonal cells. Then:

$$S\mathbf{y} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 2 \\ 5 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 2+5 \\ 2+5 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 7 \\ 7 \end{smallmatrix}\right), \qquad \mathbf{x} \cdot S\mathbf{y} = (3, 1) \cdot (7, 7) = 21 + 7 = 28$$

Equivalently $(3 + 1)(2 + 5) = 4 \cdot 7 = 28$: under $S$ every vector is reduced to its coordinate sum, and the comparison multiplies the two sums.

# In-Class Exercise 2: Answers, Item 3, The Two Raw Scores

Now the head itself, $G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$. Each score applies $G$ to the key (right slot), then dots with the query (left slot).

| | $\mathbf{x}$ **queries** $\mathbf{y}$: $\mathbf{x}^\top G\mathbf{y}$ | $\mathbf{y}$ **queries** $\mathbf{x}$: $\mathbf{y}^\top G\mathbf{x}$ |
|---|---|---|
| apply $G$ to the key | $G(2,5) = (2+2\cdot 5,\ 5) = (12,5)$ | $G(3,1) = (3+2\cdot 1,\ 1) = (5,1)$ |
| dot with the query | $(3,1)\cdot(12,5) = 36+5 = 41$ | $(2,5)\cdot(5,1) = 10+5 = 15$ |

**Average and half-difference.** $\frac{1}{2}(41+15) = 28$, the polarized value; $\frac{1}{2}(41-15) = 13$.

**Where the** $13$ **lives.** Split $G = S + A$ with $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Then $A\mathbf{y} = (5,-2)$ and $\mathbf{x}\cdot A\mathbf{y} = 15 - 2 = 13$, so

| | **raw score** | **symmetric part,** $\mathbf{x}^\top S\mathbf{y}$ | **antisymmetric part** |
|---|---|---|---|
| $\mathbf{x}$ queries $\mathbf{y}$ | $41$ | $28$ | $+13$ |
| $\mathbf{y}$ queries $\mathbf{x}$ | $15$ | $28$ | $-13$ |

Polarization used only the self-scores $16$, $49$, $121$ and recovered $28$. No self-score contains the $\pm 13$; polarization cannot see it.

# In-Class Exercise 2: Answers, Item 4, Where Symmetry Is Used

The derivation on the dot-product slide, one step per line:

| | line | what justifies it |
|---|---|---|
| (a) | $Q(\mathbf{x}+\mathbf{y}) = (\mathbf{x}+\mathbf{y})\cdot(\mathbf{x}+\mathbf{y})$ | definition of $Q$ |
| (b) | $= \mathbf{x}\cdot\mathbf{x} + \mathbf{x}\cdot\mathbf{y} + \mathbf{y}\cdot\mathbf{x} + \mathbf{y}\cdot\mathbf{y}$ | distribute: four products |
| (c) | $= Q(\mathbf{x}) + 2\,\mathbf{x}\cdot\mathbf{y} + Q(\mathbf{y})$ | $\mathbf{y}\cdot\mathbf{x} = \mathbf{x}\cdot\mathbf{y}$, so the middle two merge |
| (d) | $\mathbf{x}\cdot\mathbf{y} = \frac{1}{2}\big[Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})\big]$ | subtract and halve |

**Symmetry is used once, from (b) to (c).** Lines (a), (b), and (d) hold for any bilinear form, symmetric or not.

**Without symmetry.** With the head $G$ in place of the dot product, line (b) still holds, but its middle terms are the two different raw scores, $41$ and $15$ in item 3. They do not merge into $2\times$ one number, and subtracting and halving returns $\frac{1}{2}(41+15) = 28$, their average.

**So:** for a symmetric form, polarization returns the comparison itself; for any other form, the average of the two role assignments. Item 3 is that statement with numbers.

Part IV

# Which Distinctions Does a Learned Comparison Preserve?

eigenvectors reveal the directions; the signature counts their positive, negative, and degenerate contributions

# The Same Words Can Make Different Meanings

| | | |
|---|---|---|
| *The dog chased the cat.* | chaser: dog, | chased: cat |
| *The cat chased the dog.* | chaser: cat, | chased: dog |

The words are identical; what changes is which participant fills which role. Represent each sentence by adding fixed word vectors, and the two come out the same, because addition does not care about order:

$$\mathbf{d} + \mathbf{chase} + \mathbf{c} \;=\; \mathbf{c} + \mathbf{chase} + \mathbf{d}$$

## The compositional problem

A sentence representation must preserve *how* its parts are combined, including which participant fills which role.

Distributional word vectors supply useful lexical information: dog is near cat, far from democracy. A composition rule has to say what happens to that information when words enter a structure. Adding them throws the role assignment away.

# Attention Provides a Learned Interaction Rule

A transformer does not just add. For token $i$, one head computes (scaling, masking omitted):

$$s_{ij} = \mathbf{x}_i^\top G\, \mathbf{x}_j, \qquad \alpha_{ij} = \mathrm{softmax}_j(s_{ij}), \qquad \mathbf{h}_i = \sum_j \alpha_{ij}\, W_V \mathbf{x}_j$$

| **operation** | **contribution** |
|---|---|
| query–key score $s_{ij}$ | decides which token representations are compatible for this head |
| softmax over $j$ | converts the competing scores into weights that sum to $1$ |
| weighted value sum $\mathbf{h}_i$ | combines information into an updated representation of token $i$ |

Because $G$ need not be symmetric, $s_{ij}$ and $s_{ji}$ can differ: attention can distinguish ordered interactions, with positional and contextual information also contributing. But a learned interaction is not automatically an identifiable semantic composition rule. We want to inspect which distinctions it uses.

**The question for Part IV:** what does this head's scoring geometry distinguish, and what is it blind to?

# A Scoring Rule Can Be Blind to a Distinction

Take the symmetric half of the Session 11 head:

$$S = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top S \mathbf{y} = (x_1 + x_2)(y_1 + y_2)$$

This comparison sees only the **sum of the coordinates** of each vector. So $(3, 1)$ and $(1, 3)$, both with sum $4$, get the same score against *every* vector $\mathbf{y}$:

$$B\big((3,1), \mathbf{y}\big) = 4(y_1 + y_2) = B\big((1,3), \mathbf{y}\big)$$

| **against** $\mathbf{y} =$ | $(1, 0)$ | $(0, 1)$ | $(2, -5)$ |
|---|---|---|---|
| $B((3,1), \mathbf{y})$ | $4$ | $4$ | $-12$ |
| $B((1,3), \mathbf{y})$ | $4$ | $4$ | $-12$ |

That is stronger than the fact that their self-scores match ($16$ and $16$). **An illustration, with stipulated labels:** suppose coordinate 1 records *agent evidence* and coordinate 2 *patient evidence*. Then $(3, 1)$ and $(1, 3)$ have opposite role profiles, and this comparison cannot tell them apart. (The labels are ours for the example; nothing shows that learned coordinates mean agent and patient.)

# New Axes Show What the Comparison Keeps and What It Ignores

Describe each vector by two new coordinates instead of $x_1$, $x_2$:

$$u = \frac{x_1 + x_2}{\sqrt{2}} \text{ (shared or total component)}, \qquad v = \frac{x_1 - x_2}{\sqrt{2}} \text{ (contrast)}$$

(Dividing by $\sqrt{2}$ makes the new axes, along $(1,1)$ and $(1,-1)$, unit length.) In these coordinates

$$Q(\mathbf{x}) = (x_1 + x_2)^2 = 2u^2 + 0\,v^2$$

| | **total** $u$ | **contrast** $v$ | $Q = 2u^2 + 0\,v^2$ |
|---|---|---|---|
| $(3,1)$, more agent evidence | $4/\sqrt{2}$ | $+2/\sqrt{2}$ | $2 \cdot 8 + 0 = 16$ |
| $(1,3)$, more patient evidence | $4/\sqrt{2}$ | $-2/\sqrt{2}$ | $2 \cdot 8 + 0 = 16$ |

**The scoring rule retains the total and ignores the contrast**, and the two tokens differ only in the contrast. Coefficients $+2$ and $0$: one positive direction, no negative, one degenerate, so the signature is $(1,0,1)$.

This blindness belongs to $S$. The full head may still separate the two through its antisymmetric part $A$ ($24$ against $8$), and other heads or later stages may preserve the distinction too.

# Finding Such Axes When the Matrix Is Not Diagonal

The total and contrast axes were found by inspection. A learned comparison is handed to us in its given axes, where it usually has cross terms:

| **form** | **matrix** | **what the off-diagonal entries say** |
|---|---|---|
| the head's self-score, $(x_1 + x_2)^2$ | $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ | $\mathbf{e}_1 \cdot \mathbf{e}_2 = 1$: the given axes interact |
| Session 10's indefinite form, $x_1^2 + 4x_1x_2 + x_2^2$ | $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ | $\mathbf{e}_1 \cdot \mathbf{e}_2 = 2$: they interact more |

Along the given axes, neither form is a sum of signed squares. We need axes along which it is, so that each axis's contribution can be read separately: kept and rewarded $(+)$, kept and penalized $(-)$, or ignored $(0)$.

Rewriting in such axes, **diagonalization**, expresses the *same* comparison in coordinates that expose its separate contributions. We are neither deleting the cross terms nor changing the model's scores; every score is the same number in the new coordinates. The next slide shows such axes always exist for a symmetric matrix and how to find them; worked examples follow.

# Eigenvectors Are the Axes Along Which a Form Is Diagonal

An **eigenvector** of $S$ is a nonzero $\mathbf{v}$ that $S$ only rescales: $S\mathbf{v} = \lambda\mathbf{v}$, with $\lambda$ its **eigenvalue**. For symmetric $S$ two facts hold:

- **Eigenvectors with different eigenvalues are orthogonal.** If $S\mathbf{v}_1 = \lambda_1\mathbf{v}_1$ and $S\mathbf{v}_2 = \lambda_2\mathbf{v}_2$, then $\mathbf{v}_1^\top S\mathbf{v}_2$ equals both $\lambda_2\,\mathbf{v}_1^\top\mathbf{v}_2$ and, using $S = S^\top$, $\lambda_1\,\mathbf{v}_1^\top\mathbf{v}_2$; so $(\lambda_1 - \lambda_2)\,\mathbf{v}_1^\top\mathbf{v}_2 = 0$. (Repeated eigenvalues can be handled too; we will not need it.)
- **Along an eigenvector the form is a square.** $Q(\mathbf{v}) = \mathbf{v}^\top S\mathbf{v} = \mathbf{v}^\top(\lambda\mathbf{v}) = \lambda\,(\mathbf{v}^\top\mathbf{v})$: the eigenvalue times the ordinary squared length.

Write a vector in eigen-coordinates, $\mathbf{x} = y_1\mathbf{v}_1 + \cdots + y_n\mathbf{v}_n$ with unit eigenvectors. The cross terms vanish by the first fact, and

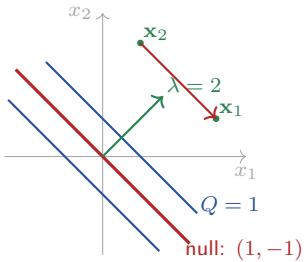
$$Q(\mathbf{x}) \;=\; \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$$

a sum of signed squares, with the eigenvalues as signs and sizes. Rescale each axis by $1/\sqrt{|\lambda_i|}$ as in Part II and the sizes go away.

## Recipe

Find the eigenvalues of the symmetric matrix. $p$ is the number of positive ones, $q$ the number of negative ones, $r$ the number of zeros.

# Worked Example: The Head's Symmetric Half



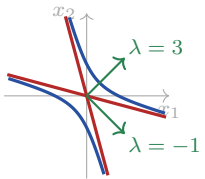

$S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Try $\mathbf{v} = (1, 1)$: $S\mathbf{v} = (2, 2) = 2\mathbf{v}$, eigenvalue 2. Try $\mathbf{v} = (1, -1)$: $S\mathbf{v} = (0, 0) = 0\mathbf{v}$, eigenvalue 0. The two are orthogonal, as promised.

In those axes, $Q = 2y_1^2 + 0\,y_2^2$: the recipe finds exactly the total and contrast axes found by inspection. One positive, one zero: signature $(1, 0, 1)$. The picture is Part II's two-lines picture, tilted so the lines are $x_1 + x_2 = \pm 1$.

The two tokens of Session 11, $\mathbf{x}_1 = (3, 1)$ and $\mathbf{x}_2 = (1, 3)$, differ by $(2, -2)$, which lies along the null direction: $Q(2, -2) = 0$. So $\mathbf{x}_1^\top S\mathbf{y} = \mathbf{x}_2^\top S\mathbf{y}$ for every key $\mathbf{y}$: the symmetric compatibility cannot tell them apart at all. Everything that does is the directional contribution, $A$.

# Worked Example: An Indefinite Form

$Q(\mathbf{x}) = x_1^2 + 4x_1x_2 + x_2^2$, matrix $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$. Try the same two directions:




| direction | $S\mathbf{v}$ | eigenvalue |
|---|---|---|
| $(1, 1)$ | $(3, 3) = 3\,\mathbf{v}$ | $3$ |
| $(1, -1)$ | $(-1, 1) = -\mathbf{v}$ | $-1$ |
| slopes $-2 \pm \sqrt{3}$ | – | $Q = 0$: null lines |

In the eigen-axes $Q = 3y_1^2 - y_2^2$; rescale $y_1$ by $\sqrt{3}$ and it is $x_1^2 - x_2^2$: signature $(1, 1, 0)$. Part II's hyperbola, tilted $45°$ and stretched, with its asymptotes as the null lines.

The eigenvalues $3$, $-1$ are not the standard hyperbola's $1$, $-1$, and the axes are not the standard axes. The signature ignores both and records only that one direction counts and one counts against.

# Worked Example: A Chain of Words and Its Level

Three words in a chain, $w_1 - w_2 - w_3$, each carrying a numerical label $x_i$ (a sentiment score, say, to be spread along the chain). The **disagreement** of a labeling is the squared difference across each link:

$$Q(\mathbf{x}) = (x_1 - x_2)^2 + (x_2 - x_3)^2 = \mathbf{x}^\top L \mathbf{x}, \qquad L = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

| **labeling** | $Q$ | **reading** |
|---|---|---|
| $(1, 2, 4)$ | $1 + 4 = 5$ | neighbors disagree by $1$ and by $2$ |
| $(3, 4, 6)$, the same plus $2$ everywhere | $1 + 4 = 5$ | the same disagreement |
| $(1, 1, 1)$ | $0$ | null: a constant labeling has no disagreement |

$L(1, 1, 1)^\top = (0, 0, 0)$, so $(1, 1, 1)$ is an eigenvector with eigenvalue $0$; the others are $1$ and $3$, along $(1, 0, -1)$ and $(1, -2, 1)$. Signature $(2, 0, 1)$.

The null axis has a plain meaning: the overall level of the labels. The disagreement measures differences and is blind to the level, which is recorded in the coordinates and invisible to the form. A null axis arises whenever a form measures only differences.

# Why the Counts Do Not Depend on the Chosen Axes

The recipe used eigenvectors, but other independent axes may also diagonalize a form: for the chain, $(1, 1, 1)$, $(1, 0, -1)$ and $(1, 1, -2)$ also do. Could a different choice give a different count? No:

## Sylvester's law of inertia (1852)

For any two choices of independent axes along which $Q$ is a sum of signed squares, the numbers of positive, negative and zero squares are the same.

**Why, for the positive count.** Suppose the first choice has $p$ positive axes and the second has $p' < p$. Let $P$ be the span of the first choice's $p$ positive axes: every nonzero vector in $P$ has $Q > 0$, a sum of positive squares. Let $N'$ be the span of the second choice's $n - p'$ non-positive axes: every vector in $N'$ has $Q \leq 0$. Count dimensions:

$$\dim P + \dim N' = p + (n - p') > n$$

Two subspaces whose dimensions add to more than $n$ share a nonzero vector $\mathbf{v}$. Then $Q(\mathbf{v}) > 0$ and $Q(\mathbf{v}) \leq 0$ at once, which is impossible. So $p' \geq p$; by symmetry $p \geq p'$. The same argument fixes $q$, and then $r = n - p - q$. This is what makes the signature a property of the form, not of the axes used to read it.

# Six Forms and Their Signatures

| **form** | **matrix** | **eigenvalues** | **signature** | **the set $Q = 1$** |
|---|---|---|---|---|
| $x_1^2 + x_2^2$ | $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ | $1,\ 1$ | $(2,0,0)$ | a circle |
| $\frac{1}{4}x_1^2 + x_2^2$ | $\left(\begin{smallmatrix} \frac{1}{4} & 0 \\ 0 & 1 \end{smallmatrix}\right)$ | $\frac{1}{4},\ 1$ | $(2,0,0)$ | an ellipse |
| $x_1^2 - x_2^2$ | $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ | $1,\ -1$ | $(1,1,0)$ | a hyperbola |
| $x_1^2 + 4x_1x_2 + x_2^2$ | $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ | $3,\ -1$ | $(1,1,0)$ | the same hyperbola, tilted |
| $(x_1 + x_2)^2$, the head | $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ | $2,\ 0$ | $(1,0,1)$ | two parallel lines, tilted |
| the chain's disagreement | $L$ | $3,\ 1,\ 0$ | $(2,0,1)$ | a cylinder about $(1,1,1)$ |

Rows one and two share a signature, and so do rows three and four: the sizes of the eigenvalues are units, and the signature forgets them, keeping only which directions count, count against, or do not count. For a matrix too large for hand work the recipe is the same and a numerical library does it; the next slide applies it to attention heads of realistic size.

# Signatures of Real Attention Heads

Any head $G = W_Q^\top W_K$ has a symmetric half with a signature $(p, q, r)$, and each count has a reading:

| | |
|---|---|
| $p$ | directions along which a token's resemblance to another *raises* the score: the head rewards likeness there |
| $q$ | directions along which resemblance *lowers* the score: the head penalizes likeness there |
| $r$ | directions the symmetric half ignores: two tokens differing only there are one token to it |

**Most of a real head is null.** With $d = 768$ and $d_k = 64$ (GPT-2 small), $G = W_Q^\top W_K$ has rank at most $64$, so $S = \frac{1}{2}(G + G^\top)$ has rank at most $128$: at least $640$ of the $768$ eigenvalues are $0$, so $r \geq 640$ and $p + q \leq 128$. Random weights give $p = q = 64$; a trained head need not be balanced.

**A tied head has $q = 0$.** With $W_Q = W_K$, $Q(\mathbf{x}) = \|W_Q \mathbf{x}\|^2 \geq 0$: no eigenvalue is negative. A tied head can reward likeness or ignore it, never penalize it; untied $W_Q$, $W_K$ are what make $q > 0$ possible.

A project-sized question: compute $(p, q, r)$ for the $144$ heads of GPT-2 small. Do positional heads and content heads have different counts?

# What a Signature Tells Us About Composition, and What It Does Not

A signature does not tell us whether a model understands who chased whom. It characterizes a scoring mechanism that composition uses.

| **what the signature gives** | **what it leaves open** |
|---|---|
| the number of directions the symmetric comparison rewards, penalizes, ignores | what those directions mean |
| the null directions: differences this comparison cannot see | whether those differences matter for meaning |
| a basis-free summary, the same in any axes | whether the antisymmetric part $A$, or another head, recovers what $S$ ignores |

To connect a direction to semantic roles, negation, modification, or scope, we must **test representations of controlled linguistic contrasts**: pairs that differ in one respect, such as *the dog chased the cat* / *the cat chased the dog*, or *the dog barked* / *the dog did not bark*. Then ask whether their difference lies in a null direction of the head.

**The method:** identify what a mechanism can distinguish, then ask whether those distinctions are the ones semantic composition requires.

# In-Class Exercise 3

1. $S = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. Check that $(1,1)$ and $(1,-1)$ are eigenvectors, find the eigenvalues, and give the signature. What is the picture of $Q = 1$?
2. $Q(\mathbf{x}) = x_1^2 - 6x_1x_2 + 9x_2^2$. Write it as a single square, write its matrix, find a null direction, and give the signature.
3. $Q(\mathbf{x}) = x_1x_2$. Write its symmetric matrix. (Careful: the off-diagonal entries are $\frac{1}{2}$ each, so that $\mathbf{x}^\top S\mathbf{x}$ gives $x_1x_2$ once.) Find the signature from the directions $(1,1)$ and $(1,-1)$.
4. $\star$ A chain of four words, $w_1 - w_2 - w_3 - w_4$, with disagreement $(x_1-x_2)^2 + (x_2-x_3)^2 + (x_3-x_4)^2$. Without computing eigenvalues, say what its signature must be and why.

Six minutes.

# In-Class Exercise 3: Answers

**1.** $S(1,1)^\top = (3,3)$, eigenvalue 3; $S(1,-1)^\top = (1,-1)$, eigenvalue 1. Signature $(2,0,0)$: an ellipse, with its long axis along $(1,-1)$ (where the form is smallest, so vectors reach $Q=1$ later).

**2.** $Q = (x_1 - 3x_2)^2$, $S = \left(\begin{smallmatrix} 1 & -3 \\ -3 & 9 \end{smallmatrix}\right)$. Null direction $(3,1)$: $Q(3,1) = (3-3)^2 = 0$. Eigenvalues 10 and 0 (trace 10, determinant 0): signature $(1,0,1)$, two parallel lines $x_1 - 3x_2 = \pm 1$.

**3.** $S = \left(\begin{smallmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{smallmatrix}\right)$. Along $(1,1)$: $Q=1$ per unit, eigenvalue $\frac{1}{2}$; along $(1,-1)$: $Q=-1$ per unit, eigenvalue $-\frac{1}{2}$. Signature $(1,1,0)$: a hyperbola whose asymptotes are the coordinate axes themselves, since $x_1 x_2 = 0$ exactly on the axes.

**4.** $\star$ The form is a sum of three squares, so no direction can have $Q<0$: $q=0$. A constant labeling $(c,c,c,c)$ is null, and nothing else is, since $Q=0$ forces every neighbor pair equal: $r=1$. So $(3,0,1)$. In general a chain of $n$ words gives $(n-1,0,1)$: one level, $n-1$ ways to disagree.

Part V

# What a Signature Does

examples, organized by the three answers

# What a Signature Decides: Overview

Three things change with the signature; each has been seen for the all-plus case. The next slides take each answer in turn.

| | **all axes $+1$** | **a $-1$ axis present** | **a $0$ axis present** |
|---|---|---|---|
| squared lengths | always positive | can be negative | zero along the axis |
| null vectors | none | a cone, where $+$ and $-$ cancel | the whole axis |
| maps preserving length | rotations and reflections: orbits stay on bounded circles | boosts: orbits run along hyperbolas, never closing | shears: orbits run along lines |
| the picture of $Q = 1$ | circle, ellipse | hyperbola | parallel lines |

The third row is the new idea in this part. A **length-preserving map** is a linear map $T$ with $Q(T\mathbf{x}) = Q(\mathbf{x})$ for every $\mathbf{x}$; for the dot product these are the rotations and reflections. Each signature has its own family, and the family is where its behavior shows: what can be done to a vector without changing its length.

# All Axes $+1$: What the Dot Product Gives You

Signature $(n, 0, 0)$: the dot product, the cosine, and every HRR, MAP and attention computation so far. The default is a declaration too, and it has consequences:

| **consequence** | **why** |
|---|---|
| no null vectors | $Q(\mathbf{x}) = \sum x_i^2 > 0$ unless $\mathbf{x} = \mathbf{0}$ |
| the cosine is a cosine | $\mathbf{x} \cdot \mathbf{y} / \sqrt{Q(\mathbf{x})\, Q(\mathbf{y})}$ lies in $[-1, 1]$: both lengths are positive, and a unit vector exists in every direction |
| length-preserving maps are rotations and reflections | a rotation moves each vector around its own circle; it never leaves that circle |
| no unbounded change | no length-preserving map takes a vector steadily away from its start: the circle is bounded |

The last row is the limitation. In an all-plus space, length-preserving change stays bounded. Whether semantics needs change that accumulates, something that makes a knife steadily better without cycling back, is what makes other signatures worth considering.

# Length-Preserving Maps in a $(+1,+1)$ Plane: What a Rotation Does

**The plane.** Both axes are declared $+1$, so the quadratic score is the familiar squared length, $Q(\mathbf{x}) = x_1^2 + x_2^2$. Both $(1,0)$ and $(0,1)$ have $Q = 1$, and all vectors with $Q = 1$ form the **unit circle**.

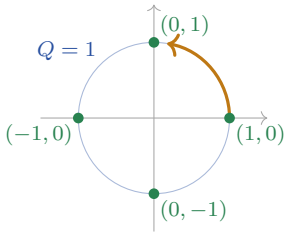
**The rotation.** Multiplying by $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ turns a vector counterclockwise through angle $\theta$. Row times column gives

$$R_\theta(x_1, x_2) = (x_1\cos\theta - x_2\sin\theta, \;\; x_1\sin\theta + x_2\cos\theta)$$

**A quarter turn.** $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$, so $R_{90^\circ} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and its rule is

$$(x_1, x_2) \longmapsto (-x_2, \; x_1)$$

Applied repeatedly: $(1,0) \to (0,1) \to (-1,0) \to (0,-1) \to (1,0)$. A longer vector: $(3,1) \to (-1,3)$, and $Q = 9 + 1 = 10$ before, $1 + 9 = 10$ after.




Four quarter turns, starting from $(1,0)$. Every point stays on the circle $Q = 1$.

# Length-Preserving Maps in a $(+1,+1)$ Plane: Length and Orbits

**The quarter turn, directly.** $Q(-x_2, x_1) = (-x_2)^2 + x_1^2 = x_1^2 + x_2^2 = Q(x_1, x_2)$.

**Any angle.** Write $c = \cos\theta$, $s = \sin\theta$, square the two new coordinates, and add:

$$
\begin{aligned}
(x_1 c - x_2 s)^2 &= x_1^2 c^2 \; - 2x_1 x_2 cs \; + x_2^2 s^2 \\
(x_1 s + x_2 c)^2 &= x_1^2 s^2 \; + 2x_1 x_2 sc \; + x_2^2 c^2 \\
Q(R_\theta \mathbf{x}) &= x_1^2(c^2 + s^2) + x_2^2(s^2 + c^2) = (c^2 + s^2)(x_1^2 + x_2^2) = Q(\mathbf{x})
\end{aligned}
$$

The cross terms cancel, and $\cos^2\theta + \sin^2\theta = 1$ finishes it.

**Orbits.** The **orbit** of a vector is the set of positions it can reach by applying the transformations under consideration. Which transformations we allow decides the orbit:

| **transformations allowed** | **orbit of** $(1,0)$ | **size** |
|---|---|---|
| every rotation $R_\theta$ | the whole unit circle | infinitely many points |
| only repeated quarter turns $R_{90^\circ}$ | $(1,0)$, $(0,1)$, $(-1,0)$, $(0,-1)$ | four points |

Either way the orbit stays on $Q = 1$: a length-preserving map moves a vector only around its own circle.

# Length-Preserving Maps in a $(+1,+1)$ Plane: Modeling *Next Season*

A season is represented by a vector; *next* by an operation on that vector. We stipulate the encoding:

| **expression** | **representation** |
|---|---|
| spring | $(1,0)$ |
| summer | $(0,1)$ |
| fall | $(-1,0)$ |
| winter | $(0,-1)$ |
| *next season* | $\mathbf{x} \mapsto R_{90^\circ}\mathbf{x}$ |

**Applying an expression is applying a function:**

$$\text{next}(\text{spring}) = R_{90^\circ}(1,0) = (0,1) = \text{summer}$$

**Composing expressions is multiplying matrices:**

$$\text{next}(\text{next}(\mathbf{x})) = R_{90^\circ}R_{90^\circ}\mathbf{x} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}\mathbf{x} = R_{180^\circ}\mathbf{x}$$

**Four applications return the start:** $R_{90^\circ}^4 = R_{360^\circ} = I$, the **identity map**, which leaves every vector unchanged.

**Reading the identity.** *Four seasons from now* returns the same *season*, not the same time: the encoding tracks position in the cycle, not the year.

**Signature versus choice.** The positive-definite plane supports circular rotations; choosing a quarter turn gives the four-season cycle. The signature permits the geometry; our encoding and operation establish the semantic cycle.

**Payoff:** we represent both the things we talk about and a reusable operation on them, and the representation makes the operation's cyclic behavior exact.

# Length-Preserving Maps in a $(+1,-1)$ Plane: Boosts

Replace $\cos$, $\sin$ by $\cosh$, $\sinh$, the even and odd halves of $e^{\varphi}$:

$$B_\varphi = \begin{pmatrix} \cosh\varphi & \sinh\varphi \\ \sinh\varphi & \cosh\varphi \end{pmatrix}, \qquad B_\varphi \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cosh\varphi \\ \sinh\varphi \end{pmatrix}$$

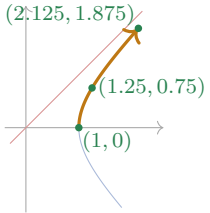
It preserves $x_1^2 - x_2^2$ because $\cosh^2\varphi - \sinh^2\varphi = 1$: $Q(B_\varphi \mathbf{x}) = (\cosh^2\varphi - \sinh^2\varphi)(x_1^2 - x_2^2)$. The orbit of $(1,0)$ is the hyperbola, and no $\varphi \neq 0$ brings the vector back: $\cosh\varphi \geq 1$ grows without bound.

**Numbers**, with $\varphi = \ln 2$, so $\cosh\varphi = 1.25$, $\sinh\varphi = 0.75$, and $Q = 1$ throughout:

$(1,0) \to (1.25,\ 0.75) \to (2.125,\ 1.875) \to \cdots$

**Null vectors stay null**: $B_\varphi(1,1)^\top = (2,2)^\top$, on the same diagonal, stretched by $2$.

The map is a **boost**; in physics, with $Q = t^2 - x^2$, it is a change of velocity.




The orbit of $(1,0)$ under $B_\varphi$ is the hyperbola $Q = 1$ of Part II; the red diagonal is its asymptote.

# Repeated Boosts in a $(+1,-1)$ Plane Never Cycle Back

**Step 1: boosts compose by adding**, $B_\alpha B_\beta = B_{\alpha+\beta}$. So $B_\varphi^n = B_{n\varphi}$ takes $(1,0)$ to $(\cosh n\varphi,\ \sinh n\varphi)$; with $\varphi = \ln 2$, that is $\left(\frac{1}{2}(2^n + 2^{-n}),\ \frac{1}{2}(2^n - 2^{-n})\right)$.

**Step 2: compare with a quarter turn**, both from $(1,0)$:

| $n$ | **quarter turn** $R_{90^\circ}^n$ | **boost** $B_{\ln 2}^n$ | **boost:** $x^2 - y^2$ |
|---|---|---|---|
| 0 | $(1,0)$ | $(1,\ 0)$ | $1$ |
| 1 | $(0,1)$ | $(1.25,\ 0.75)$ | $1.5625 - 0.5625 = 1$ |
| 2 | $(-1,0)$ | $(2.125,\ 1.875)$ | $4.515625 - 3.515625 = 1$ |
| 3 | $(0,-1)$ | $(4.0625,\ 3.9375)$ | $16.50390625 - 15.50390625 = 1$ |
| 4 | $(1,0)$, back | $(8.03125,\ 7.96875)$ | $1$ |

**Step 3: why the boost never returns.** For $\varphi, n > 0$, $\sinh(n\varphi) > 0$: the second coordinate never returns to $0$, so neither does the vector to $(1,0)$.

**Never cycling is not the same as irreversible.** $B_{-\varphi}B_\varphi = I$: $B_{-\ln 2} = \left(\begin{smallmatrix} 1.25 & -0.75 \\ -0.75 & 1.25 \end{smallmatrix}\right)$ takes $(1.25, 0.75)$ to $(1.5625 - 0.5625,\ -0.9375 + 0.9375) = (1,0)$.

Keep doing a quarter turn and you return to the start. Keep doing a positive boost and you do not; to return, reverse the operation.

# Length-Preserving Maps in a $(+1, 0)$ Plane: Shears

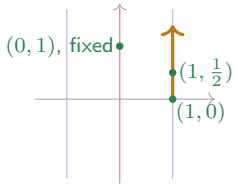
With $Q = x_1^2$, a map preserves length if it leaves $x_1$ alone. The simplest adds a multiple of $x_1$ to the null coordinate:

$$H_\theta = \begin{pmatrix} 1 & 0 \\ \theta & 1 \end{pmatrix}, \qquad H_\theta \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 + \theta x_1 \end{pmatrix}$$

$Q$ is preserved because the first coordinate is untouched. The orbit of $(1, 0)$ is the line $x_1 = 1$: the vector slides along the null direction at a steady rate $\theta$, and neither returns nor runs away.

**Numbers.** $H_{1/2}$: $(1, 0) \to (1, \frac{1}{2})$; $(3, 1) \to (3, 2\frac{1}{2})$, with $Q = 9$ before and after; $(0, 1) \to (0, 1)$: a vector on the null axis is fixed.

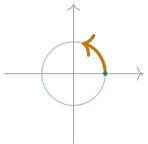
This map is a **shear**. It leaves the null axis alone and changes the null coordinate of every other vector in proportion to its $x_1$. The null coordinate is a record that other coordinates can write to, and that length never reads.



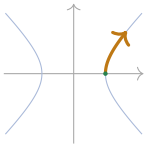

The orbit of $(1, 0)$ under $H_\theta$ is the line $Q = 1$ of Part II; the red line is the null axis.

# Each Orbit Stays Within Its Level Set

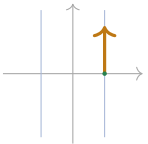
The three previous slides side by side. The orange path is the **orbit** of $(1, 0)$, the positions it reaches under the length-preserving maps; the pale blue curve is the **level set** $Q = 1$ drawn in Part II:



$(+1, +1)$: rotations $R_\theta$



$(+1, -1)$: boosts $B_\varphi$



$(+1,\ 0)$: shears $H_\theta$

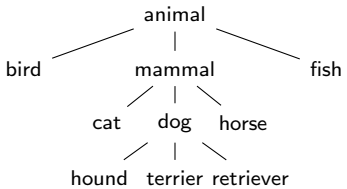
The orbit stays within the level set. A map that preserves $Q$ keeps a vector with $Q = 1$ among the vectors with $Q = 1$: preservation guarantees containment. It need not reach all of the set: rotations cover the whole circle, but boosts stay on one branch of the hyperbola, and shears stay on one of the two lines.

Choosing a signature chooses the family of maps: bounded motion around a circle; accumulating motion along a hyperbola; steady sliding along a line.

**The examples that follow** each ask a semantic question first (what should change, what should stay fixed, what should happen on repetition), then give the geometry that answers it.

# A Taxonomy Needs Room for Branching

A fragment of a noun hierarchy:




An embedding should preserve two things:

- descendants become **more specific**: further from the root;
- different branches remain **distinguishable**: siblings not crowded together.

**Counting.** If every node has three children, depth $d$ holds $3^d$ nodes. Place depth $d$ on a ring of radius $d$ around the root and compare the room on the ring:

| **depth** $d$ | **nodes** $3^d$ | **plane** $2\pi d$ | **hyperbolic** $2\pi \sinh d$ |
|---|---|---|---|
| 1 | 3 | 6.3 | 7.4 |
| 2 | 9 | 12.6 | 22.8 |
| 3 | 27 | 18.8 | 62.9 |
| 4 | 81 | 25.1 | 171.5 |

In the plane the ring grows in step with $d$; the tree grows by a factor of 3 per level. By depth 4, 81 nodes share a ring of length 25: siblings crowd.

The motivating issue is room for branching. The next slides build a space that has it.

# Hyperbolic Geometry for a Taxonomy: The Surface

**The construction.** Use an ambient space with one negative direction, $Q(\mathbf{x}) = -x_0^2 + x_1^2 + x_2^2$, and restrict representations to one surface in it:

$$-x_0^2 + x_1^2 + x_2^2 = -1, \qquad x_0 > 0$$

Distances measured on that surface give **hyperbolic geometry**:

$$d(\mathbf{x}, \mathbf{y}) = \operatorname{arcosh}(-\mathbf{x} \cdot \mathbf{y}), \qquad \mathbf{x} \cdot \mathbf{y} = -x_0 y_0 + x_1 y_1 + x_2 y_2$$

with $\operatorname{arcosh}$ the inverse of $\cosh$. The surface is the hyperbola of Part II with the $-1$ axis vertical, spun into a bowl.

| **point** | **coordinates** | **distance** |
|---|---|---|
| $\mathbf{o}$, the root | $(1,\ 0,\ 0)$ | – |
| $\mathbf{p}$, depth 2 | $(\cosh 2,\ \sinh 2,\ 0) \approx (3.76,\ 3.63,\ 0)$ | $\mathbf{o} \cdot \mathbf{p} = -\cosh 2$, so $d(\mathbf{o}, \mathbf{p}) = 2$ |
| $\mathbf{q}$, opposite branch | $(\cosh 2,\ -\sinh 2,\ 0)$ | $\mathbf{p} \cdot \mathbf{q} = -\cosh 4$, so $d(\mathbf{p}, \mathbf{q}) = 4$ |
| $\mathbf{r}$, at right angles | $(\cosh 2,\ 0,\ \sinh 2)$ | $\mathbf{p} \cdot \mathbf{r} = -\cosh^2 2$, so $d(\mathbf{p}, \mathbf{r}) \approx 3.34$ |

Every point has $Q = -1$: for $\mathbf{p}$, $-\cosh^2 2 + \sinh^2 2 = -1$. $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ all lie at depth 2: distance 2 from the root.

# Hyperbolic Geometry for a Taxonomy: Reading the Coordinates

| **taxonomy** | **on the surface** |
|---|---|
| the root, entity or animal | a chosen point $\mathbf{o} = (1, 0, 0)$ |
| depth, how specific | distance from $\mathbf{o}$: $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ all at depth $2$ |
| which branch | the direction of $(x_1, x_2)$ |
| siblings stay apart | two depth-2 nodes at right angles are $3.34$ apart, against $2\sqrt{2} \approx 2.83$ in the plane |

**Two misreadings to avoid.** $x_0$ is *not* generality, and the negative sign does *not* mean semantic negation. The negative axis alone does not establish the encoding: the surface constraint and its metric are essential.

**The precise comparison.** At fixed dimension, Euclidean space has *polynomial* volume growth: more dimensions raise the power, never to an exponential. Hyperbolic space has *exponential* volume growth, matching the growth pattern of a branching tree.

In NLP: Nickel and Kiela (2018) embed the WordNet noun hierarchy on this surface in a handful of dimensions, entity near the root and chihuahua far down the side.

# Degree Modification: Does Repeated *Better* Return to the Start?

Increasing functional quality:

knife, adequate for cutting $\longrightarrow$ better for cutting $\longrightarrow$ better still $\longrightarrow \cdots$

Two operations: **selecting the relevant dimension** (for a knife, cutting, its purpose or telic role), and **increasing a degree on that dimension** (each *better* moves further along it).

| **question about the phenomenon** | **expected answer** |
|---|---|
| does repeated *better* return to the start? | no: *better still* is never again merely adequate |
| can a step be undone? | yes: *worse* undoes *better* |
| do steps combine? | a little better, then a little more, is better by the sum |

**What an all-plus plane offers.** If each step must preserve a declared length, then with both axes $+1$ it is a rotation. Repeated, it stays on a bounded circle: it returns *exactly* after finitely many steps only when its angle is a rational multiple of $2\pi$, and otherwise circles without arriving back. Either way the degree cannot keep growing.

**A qualification.** Rotations preserve Euclidean length; arbitrary Euclidean semantic functions need not.

Which length-preserving map fits the three answers?

# Degree Modification as a Boost: The Encoding and the Numbers

Functional Geometric Algebra (Pustejovsky 2026) proposes this encoding. Give knife two coordinates $(f, t)$:

- a *form* axis $f$, declared $+1$;
- the selected evaluative scale $t$, here cutting, declared $-1$.

The baseline is $(1, 0)$: **zero displacement on the selected evaluative scale**. A knife already has a telic role; modification need not create it. Each step of *better* is the boost $B_\varphi$ with $\varphi = \ln 2$, so $\cosh \varphi = 1.25$, $\sinh \varphi = 0.75$:

| **step** | **coordinates** $(f, t)$ | **preserved quantity** $f^2 - t^2$ |
|---|---|---|
| baseline | $(1,\ 0)$ | $1 - 0 = 1$ |
| one increment | $(1.25,\ 0.75)$ | $1.5625 - 0.5625 = 1$ |
| two increments | $(2.125,\ 1.875)$ | $4.515625 - 3.515625 = 1$ |

## The gloss

Both coordinates can grow while their squared difference stays fixed. A circle cannot accommodate that unbounded growth at fixed radius; a hyperbola can.

# Degree Modification as a Boost: Degree, Combination, Reversal

**The degree parameter.** Every point on this branch of the hyperbola is $(f, t) = (\cosh\varphi, \sinh\varphi)$, and $\varphi$ is the proposed degree: $0$ at the baseline, $\ln 2$ after one step, $2\ln 2$ after two.

Each answer on the phenomenon slide becomes a property of boosts:

| **expected answer** | **property** | **check** |
|---|---|---|
| steps combine by adding | $B_\alpha B_\beta = B_{\alpha+\beta}$ | $\cosh(2\ln 2) = \frac{1}{2}(4 + \frac{1}{4}) = 2.125$, $\sinh(2\ln 2) = \frac{1}{2}(4 - \frac{1}{4}) = 1.875$: the third row |
| a step can be undone | $B_{-\varphi} B_\varphi = I$ | *worse* by $\ln 2$ takes $(1.25, 0.75)$ back to $(1, 0)$ |
| repetition never returns | $\cosh\varphi > 1$ for $\varphi \neq 0$ | $f$ grows with every positive step: nonperiodic, never wrapping around |

Two positive increments produce a larger increment. That is the compositional claim, made explicit.

Status: a proposed implementation of degree-sensitive modification, not a consequence of the meaning of good or better. It predicts the three answers, and it is testable against data.

# A Zero Direction: What Can Change Without Changing This Score?

A short discourse about one referent:

*A dog barked.* *It ran off.* *Later it came back.*

Keep a simple counting record for the referent: $(1, n)$, where the first coordinate is its content (held fixed here) and $n$ is a **discourse index**, counting the updates since it was introduced. Declare the content axis $+1$ and the index axis $0$, so $Q(1, n) = 1^2 + 0 \cdot n^2 = 1$ for every $n$.

An update is the shear $H_a(x_1, x_2) = (x_1,\ x_2 + a\,x_1)$, which on a record gives $H_a(1, n) = (1,\ n + a)$. With $a = 1$:

| **sentence** | **record** | **index** $n$ | $Q$ |
|---|---|---|---|
| *A dog barked.* | $(1, 0)$ | $0$ | $1$ |
| *It ran off.* | $(1, 1)$ | $1$ | $1$ |
| *Later it came back.* | $(1, 2)$ | $2$ | $1$ |

The record changes, but this quadratic score ignores the index coordinate. The zero direction is not meaningless; it is invisible to this chosen form. The orbit is the line $x_1 = 1$, one of the two lines $Q = 1$ of Part II.

Illustrative encoding of an additive update, not a model of discourse.

# Zero Directions Already Seen in This Deck

Two null axes have appeared in this deck already, neither by declaration:

| **where** | **the null direction** | **what it carries** |
|---|---|---|
| the chain of words (Part IV) | $(1, 1, 1)$: the same label everywhere | the **level** of the labels, which the disagreement cannot measure |
| the attention head (Part IV) | $(1, -1)$: the difference between the two tokens | everything the symmetric half cannot see about them; the direction of attention lives in $A$ |

In both, the null coordinate is recorded but not counted: it is present in the vector, every length and every similarity ignore it, and a shear is the kind of map that can change it without any length noticing, as the discourse index just did.

That makes a $0$ axis a side channel. Information put there survives, but nothing built on length or similarity can retrieve it; only an operation that reads coordinates directly can. The next slide shows one such operation, once vectors multiply; the slide after it is why FGA keeps $0$ axes out of the meaning space.

# Example of a $0$ Axis, Once Vectors Multiply: Dual Numbers

Part III showed that if vectors multiply with $\mathbf{v}\mathbf{v} = Q(\mathbf{v})$, then an axis obeys $\mathbf{e}\mathbf{e} = \mathbf{e} \cdot \mathbf{e}$. Take a single axis declared $0$ and call it $\varepsilon$: the one rule is $\varepsilon^2 = 0$, and numbers of the form $a + b\varepsilon$ multiply by ordinary algebra plus that rule:

$$(a + b\varepsilon)(c + d\varepsilon) = ac + (ad + bc)\,\varepsilon + bd\,\varepsilon^2 = ac + (ad + bc)\,\varepsilon$$

These are the **dual numbers**. Cube $2 + \varepsilon$:

$$(2 + \varepsilon)^2 = 4 + 4\varepsilon, \qquad (4 + 4\varepsilon)(2 + \varepsilon) = 8 + 4\varepsilon + 8\varepsilon = 8 + 12\varepsilon$$

The number part is $2^3 = 8$. The $\varepsilon$ part is $12 = 3 \cdot 2^2$, the derivative of $x^3$ at $x = 2$. This is general: for any polynomial, $f(x + \varepsilon) = f(x) + f'(x)\,\varepsilon$, because every $\varepsilon^2$ term vanishes and what survives is exactly the first-order change. Automatic differentiation in machine-learning libraries is this computation.

So the null coordinate carries a **rate**: a value and the direction it is moving, in one object. A sense written as $\mathbf{w} + \varepsilon\mathbf{v}$ is a word vector together with its drift, and every similarity reads $\mathbf{w}$ alone while the drift rides along. Session 13a builds the product this slide assumed.

# Mixed Signatures: FGA's Design Table

With the three answers understood, here is how FGA proposes to spend them, with the reasons this part has supplied:

| axis | proposed use | what the declaration buys | what it costs |
|---|---|---|---|
| $+1$ | spatial, extensional, similarity-preserving meaning | the cosine; no null vectors | length-preserving change stays bounded |
| $-1$ | intensional, modal, contrastive meaning | boosts: unbounded, nonperiodic, reversible change | null vectors; lengths that can be negative |
| $0$ | not used for meaning | a side channel: a rate or a level | invisible to every similarity |

The paper's first application is all plus, with unit entity vectors, so that similarity between entities is their cosine; everything in the bridge deck assumed this. The $-1$ axes are a proposal for where that assumption should be given up, and $0$ axes are kept out because nothing stored on one can be found again by similarity.

A signature, on this view, is a hypothesis about which operations a meaning space has, stated in advance and checkable: a boost that is supposed to model good either fits the data or does not.

# Three Constraints on Semantic Change

Each signature answered a semantic question: what should change, what should remain fixed, what should happen on repetition.

| **intended behavior** | **example** | **geometry** | **what remains fixed** |
|---|---|---|---|
| return through a cycle | *next season*, four times | circle | $x^2 + y^2$ |
| accumulate a nonperiodic degree change | *better for cutting*, repeated | hyperbola | $x^2 - y^2$ |
| update an unscored coordinate | a discourse index | line | $x^2$ |

The taxonomy used a negative direction for a different purpose: not a kind of change, but **room for branching**, through a surface and its metric.

## The lesson of Part V

We choose a geometry because it supports a desired pattern of composition, then test whether that pattern fits the linguistic phenomenon. The signature permits the geometry; the encoding and operation make the claim.

# What Each Answer Does: Details

| answer | $Q = 1$ is | null vectors | length-preserving maps | seen in | once vectors multiply |
|---|---|---|---|---|---|
| $+1$ | a circle | none | rotations: bounded, around a circle | the cosine; word vectors; HRR, MAP; FGA's entities | $\mathbf{e}^2 = +1$ |
| $-1$ | a hyperbola | a cone, where $+$ and $-$ cancel | boosts: nonperiodic, reversible | hyperbolic embeddings of WordNet; FGA's better knife | $\mathbf{e}^2 = -1$ |
| $0$ | two lines | the axis itself | shears: steady sliding | the chain's level; a head's null directions; a discourse index; autodiff | $\varepsilon^2 = 0$, dual numbers |

Read down the first column: one question, three answers. Read across any row: one answer, followed into a picture, a set of null vectors, a family of maps, and the places it has appeared. The signature of a space is this table, applied once per axis.

# In-Class Exercise 4

1. Apply the boost $B_{\ln 2}$, with $\cosh = 1.25$ and $\sinh = 0.75$, to $(3, 1)$. Check that $Q = x_1^2 - x_2^2$ is unchanged. Then apply it to the null vector $(1, -1)$: what happens?
2. In the seasons plane, spring $= (1, 0)$ and winter $= (0, -1)$. Which rotation takes winter to spring? How many times must *next season* be applied to winter to reach fall?
3. Under $(+1, 0)$, apply the shear $H_2$ to $(1, 0)$, to $(2, 3)$, and to $(0, 5)$. Which vectors does a shear leave fixed, and why?
4. $\star$ Suppose the telic axis of knife were declared $0$ rather than $-1$, so that *better* had to be a shear. Describe what repeated *better* would do, and say what a *very good knife* and a *knife* would have in common that the boost version denies them.

Six minutes.

# In-Class Exercise 4: Answers

**1.** $B(3,1)^\top = (1.25 \cdot 3 + 0.75 \cdot 1,\ 0.75 \cdot 3 + 1.25 \cdot 1) = (4.5,\ 3.5)$. $Q$ before: $9 - 1 = 8$; after: $20.25 - 12.25 = 8$. The null vector: $B(1,-1)^\top = (0.5,\ -0.5)$, still on the other diagonal, shrunk by 2. A boost stretches one null line and shrinks the other by the same factor.

**2.** winter $= (0,-1)$ to spring $= (1,0)$ is a quarter turn counterclockwise, $R_{90^\circ}$, which is *next season*. From winter, *next season* applied three times: spring, summer, fall.

**3.** $H_2(1,0)^\top = (1,2)$; $H_2(2,3)^\top = (2,7)$; $H_2(0,5)^\top = (0,5)$. A shear adds $\theta x_1$ to the second coordinate, so it fixes exactly the vectors with $x_1 = 0$: the null axis. Everything else slides, by an amount proportional to its $x_1$.

**4.** $\star$ With the telic axis null, *better* would be $H_\theta$: $(1,0) \to (1,\theta) \to (1,2\theta) \to \cdots$. The telic coordinate grows steadily and the form coordinate never changes, so a *very good knife* and a *knife* would have the same form component and, since the telic axis is invisible to similarity, the same similarity to everything. The boost version makes the form component grow too and keeps the telic part visible to comparison. Which is right is an empirical question about how *better* behaves; the signature states the two hypotheses precisely.

# Part VI
# Where It Goes Next

the signature as a rule for multiplying vectors

# The Signature Becomes a Multiplication Rule

Part III derived, from the single assumption $\mathbf{v}\mathbf{v} = Q(\mathbf{v})$, that $\mathbf{u}\mathbf{v} + \mathbf{v}\mathbf{u} = 2\,\mathbf{u}\cdot\mathbf{v}$. For the axes themselves this gives a complete multiplication table:

| rule | status | what it does to a product of axes |
|---|---|---|
| $\mathbf{e}_i\mathbf{e}_i = +1,\ -1,$ or $0$ | declared | **collapse**: equal neighbors become the declared sign |
| $\mathbf{e}_i\mathbf{e}_j = -\,\mathbf{e}_j\mathbf{e}_i$ for $i \neq j$ | derived, Part III | **swap**: different neighbors trade places, with a sign |

The swap rule never consults the declaration; the collapse rule is the declaration. A computation whose equal neighbors all cancel is the same under every signature; one that collapses a pair depends on it:

| from the bridge deck | formula | collapses a pair? | depends on the signature? |
|---|---|---|---|
| the inner product | $\frac{1}{2}(\mathbf{u}\mathbf{v} + \mathbf{v}\mathbf{u})$ | yes | yes |
| the wedge | $\frac{1}{2}(\mathbf{u}\mathbf{v} - \mathbf{v}\mathbf{u})$ | no: the $\mathbf{e}_i\mathbf{e}_i$ terms cancel | no |

This is why the bridge deck could compute $\mathbf{u}\wedge\mathbf{v}$ in two dimensions without saying what $\mathbf{e}_1\mathbf{e}_1$ was. 13a builds the product from here; 13c uses the second row for binding.

# One Axis, Three Algebras

With a single axis $\mathbf{e}$, the elements are $a + b\mathbf{e}$, and the product has exactly one place where the declaration enters:

$$(a + b\mathbf{e})(c + d\mathbf{e}) \;=\; \big(ac + bd\,\mathbf{ee}\big) + (ad + bc)\,\mathbf{e}$$

| $\mathbf{ee}$ | $(2 + 3\mathbf{e})(1 + 4\mathbf{e})$ | $(2 + 3\mathbf{e})^2$ | **known as** |
|---|---|---|---|
| $+1$ | $2 + 12 + 11\mathbf{e} = 14 + 11\mathbf{e}$ | $4 + 9 + 12\mathbf{e} = 13 + 12\mathbf{e}$ | the split-complex numbers |
| $-1$ | $2 - 12 + 11\mathbf{e} = -10 + 11\mathbf{e}$ | $4 - 9 + 12\mathbf{e} = -5 + 12\mathbf{e}$ | the complex numbers |
| $0$ | $2 + 0 + 11\mathbf{e} = 2 + 11\mathbf{e}$ | $4 + 0 + 12\mathbf{e} = 4 + 12\mathbf{e}$ | the dual numbers (Part V) |

The $\mathbf{e}$-coefficients, $11$ and $12$, are the same in every row: $ad + bc$ never meets $\mathbf{ee}$. Only the number part consulted the declaration, through the one term $bd\,\mathbf{ee}$.

**The complex numbers are the middle row.** There is no mysterious $i$: one axis was declared $-1$, and $-9$ is that declaration showing up inside a computation. The three length-preserving maps of Part V reappear here as multiplication by $\cos\theta + \mathbf{e}\sin\theta$, by $\cosh\varphi + \mathbf{e}\sinh\varphi$, and by $1 + \theta\mathbf{e}$. Session 13a and 13b take this up.

# Summary

1. **Every inner product has a matrix in the middle**, and every such matrix has a signature. The dot product's is $(n, 0, 0)$; you have been using one all along without choosing it.
2. **A signature is the answer to one question per axis**: is the axis's square $+1$, $-1$, or $0$? Sizes can be rescaled away; signs cannot. The triple $(p, q, r)$ counts the answers.
3. **Lengths can be negative or zero.** A null vector is nonzero and invisible to length; a $0$ axis is invisible to every comparison. The sets $Q = 1$ are a circle, a hyperbola, two lines.
4. **Polarization**, $\mathbf{x} \cdot \mathbf{y} = \frac{1}{2}[Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})]$, lets quadratic scores determine symmetric comparisons, keeps only the symmetric half of a matrix, and, once vectors multiply, gives $\mathbf{uv} + \mathbf{vu} = 2\,\mathbf{u} \cdot \mathbf{v}$.
5. **To read a signature off a form**, find the eigenvalues of its symmetric matrix and count signs. Sylvester's law says the count does not depend on the axes. The head's $S$ is $(1, 0, 1)$; a word chain's disagreement is $(n-1, 0, 1)$.
6. **What a signature does** shows in its length-preserving maps: rotations for $(+1, +1)$, which stay bounded; boosts for $(+1, -1)$, which accumulate without cycling; shears for $(+1, 0)$, which slide. Hyperbolic geometry, built with a $-1$ axis, gives taxonomies room to branch; FGA proposes $-1$ for degree modification; a $0$ axis holds what a form does not score.

# Exercises

1. Under $(+1,+1,-1)$, find three null vectors that are not multiples of one another, and describe the set $Q=0$ in words.
2. Find the signature of $Q(\mathbf{x})=2x_1x_2+2x_2x_3$ on three axes. (Write the symmetric matrix; one eigenvalue is $0$, and the other two are $\pm\sqrt{2}$.)
3. Polarize $Q(\mathbf{x})=\mathbf{x}^{\top}G\mathbf{x}$ for $G=\left(\begin{smallmatrix}2&6\\0&1\end{smallmatrix}\right)$ by hand, in general $\mathbf{x}$ and $\mathbf{y}$, and confirm that the result is $\mathbf{x}^{\top}S\mathbf{y}$ with $S=\tfrac{1}{2}(G+G^{\top})$.
4. Show that $B_{\varphi}B_{\psi}=B_{\varphi+\psi}$ for boosts, using $\cosh(\varphi+\psi)=\cosh\varphi\cosh\psi+\sinh\varphi\sinh\psi$ and the matching identity for $\sinh$. Conclude that good applied twice is a single, larger boost.
5. Write the matrix of a general length-preserving map for $(+1,0)$ in the plane. (It must fix $x_1$ up to sign; what can it do to $x_2$?) Which of them are shears?
6. $\star$ Take a small transformer you can run and, for a few heads, compute $(p,q,r)$ of the symmetric half of $W_Q^{\top}W_K$. Report whether any head has $q$ close to $0$, and what such a head attends to.

# Reading

- J. J. Sylvester, *A demonstration of the theorem that every homogeneous quadratic polynomial is reducible by real orthogonal substitutions to the form of a sum of positive and negative squares*, Philosophical Magazine 4, 1852 – the law of inertia, in its original form.
- G. Strang, *Introduction to Linear Algebra*, 5th ed., Wellesley-Cambridge, 2016 – Section 6.5 on positive definite matrices and the signs of eigenvalues.
- L. Dorst, D. Fontijne and S. Mann, *Geometric Algebra for Computer Science*, Morgan Kaufmann, 2007 – Chapter 3, metric products of subspaces, for what the next sessions build on a signature.
- M. Nickel and D. Kiela, *Learning continuous hierarchies in the Lorentz model of hyperbolic geometry*, ICML 2018 – the $-1$ axis in use for a taxonomy.
- J. Pustejovsky, *Toward a Functional Geometric Algebra for natural language semantics*, arXiv:2604.25902v2, 2026 – the signature as a design parameter; the good knife example.
- Session 13a for the Clifford product built from the declaration; 13b for the one-axis algebras as programs; 13c for the wedge, the dual, and binding.